



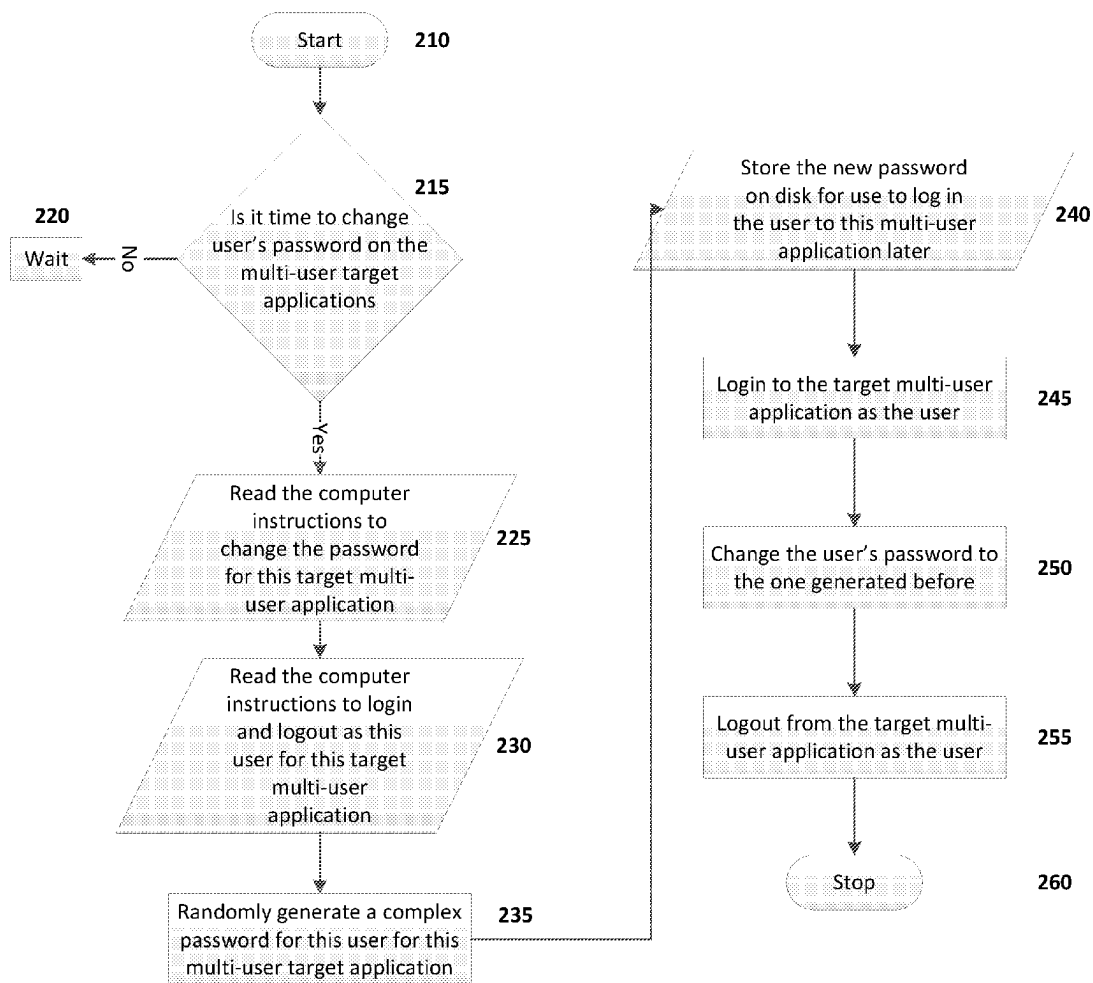
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(19) **United States**(12) **Patent Application Publication**
Kapoor(10) **Pub. No.: US 2016/0373436 A1**(43) **Pub. Date: Dec. 22, 2016**(54) **SECURED APPLICATION ACCESS SYSTEM
AND METHOD WITH FREQUENTLY
CHANGING PASSWORDS**(71) Applicant: **Rohit Kapoor**, Tokyo (JP)(72) Inventor: **Rohit Kapoor**, Tokyo (JP)(21) Appl. No.: **15/169,541**(22) Filed: **May 31, 2016****Related U.S. Application Data**

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(2013.01); **H04L 63/10** (2013.01); **G06F**
2221/2131 (2013.01)(57) **ABSTRACT**

A system and method for automatically and frequently changing or updating a user password without manual intervention and without the knowledge of the user while accessing at least one target multi-user computer application stored on a target multi-user computer applications server through a login management server over a first computer network. A changed password is stored provided to the user when demanded.



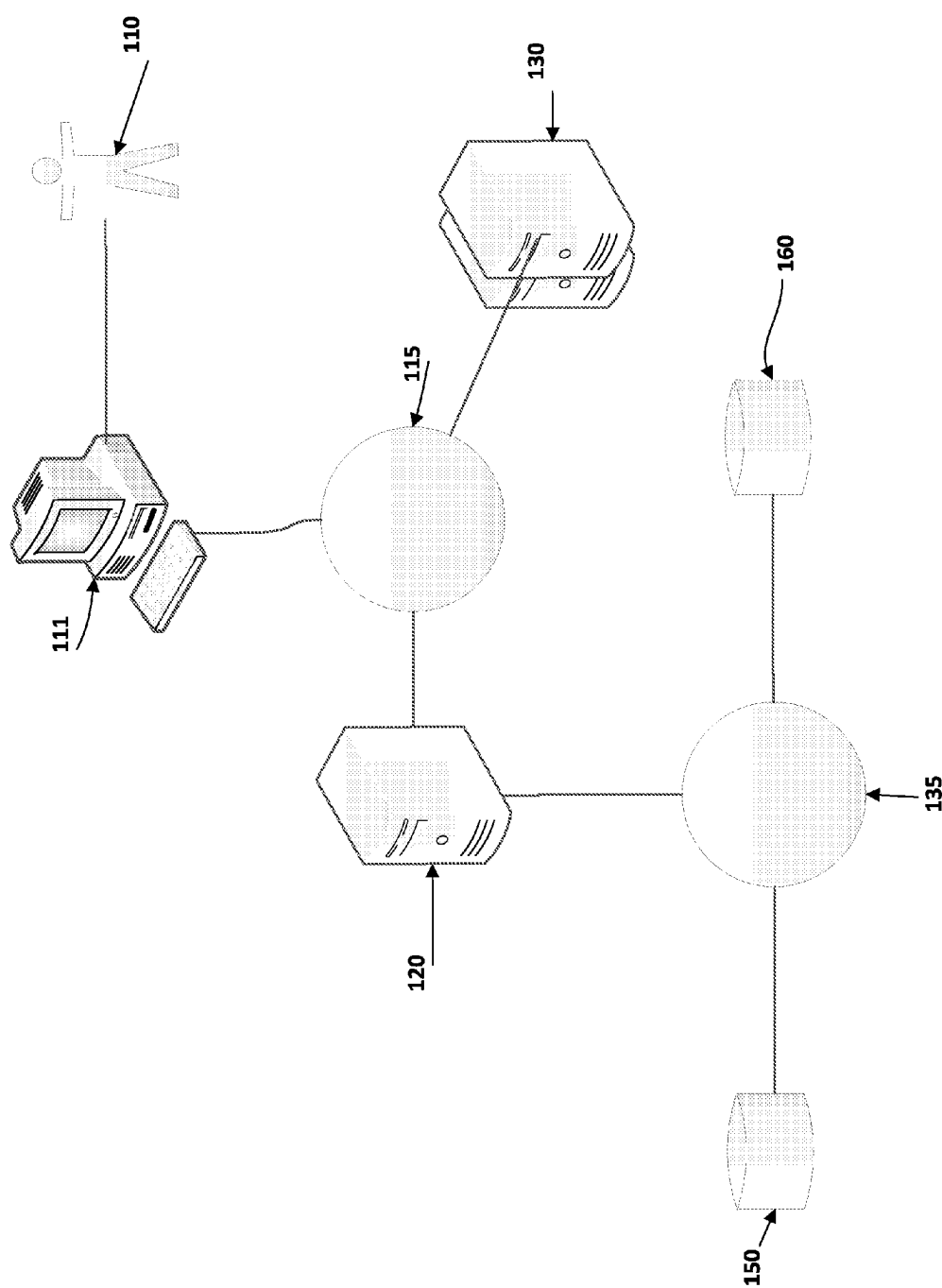


FIG. 1

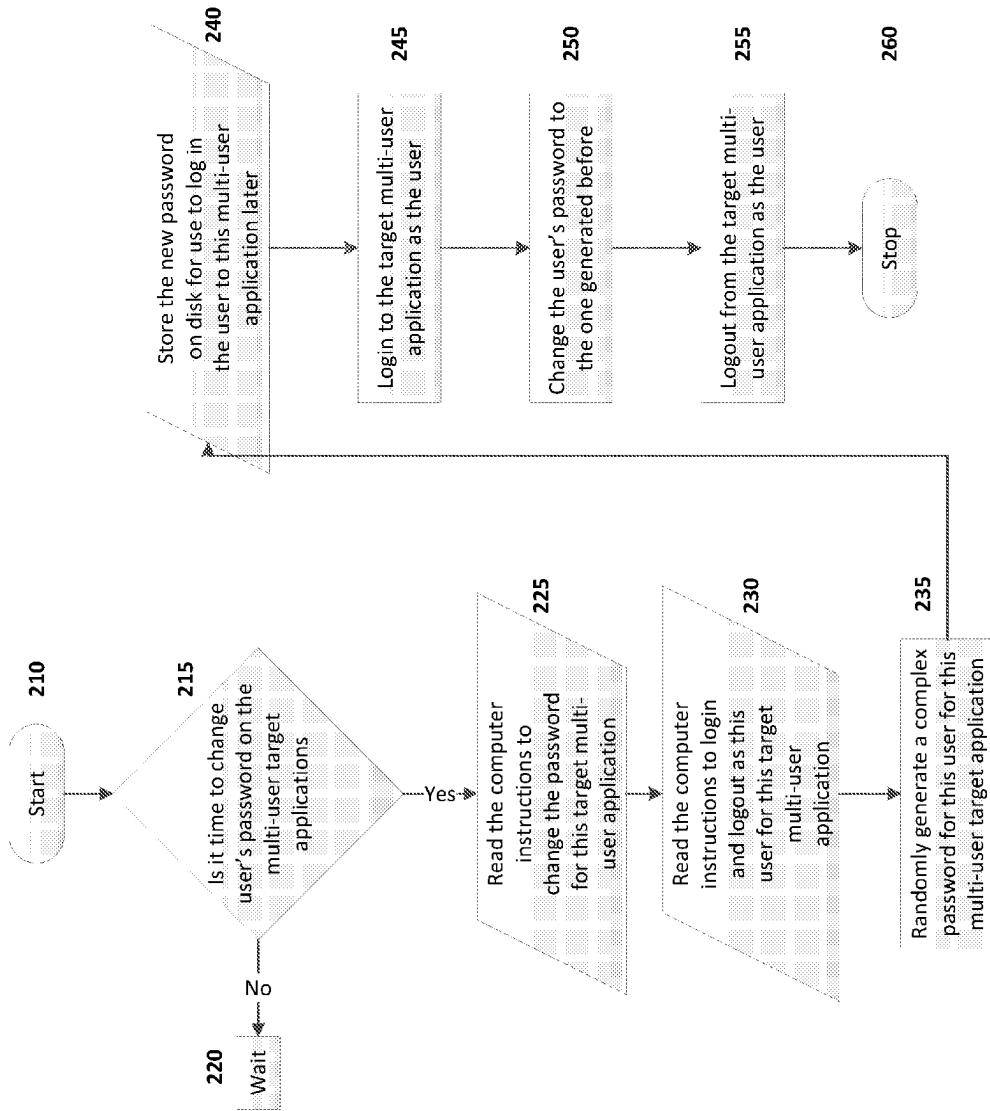


FIG. 2

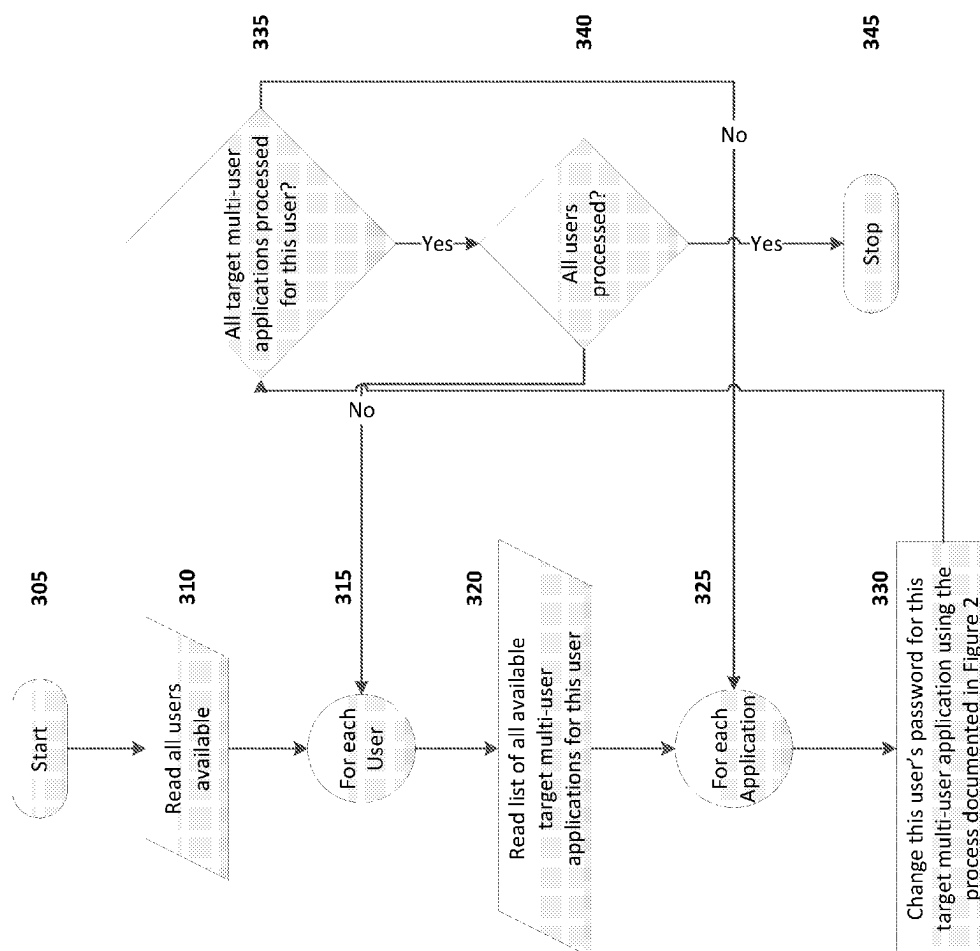


FIG. 3

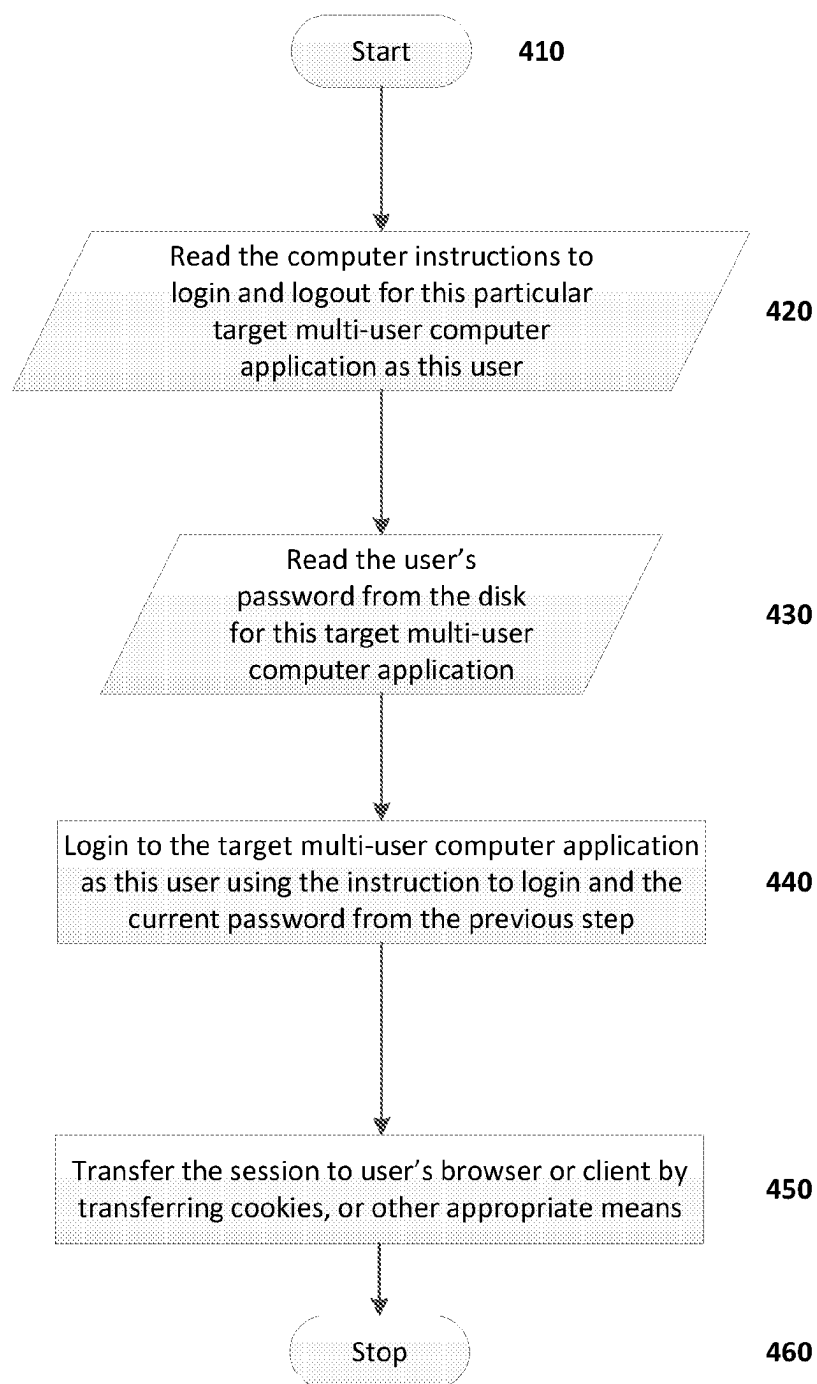


FIG. 4

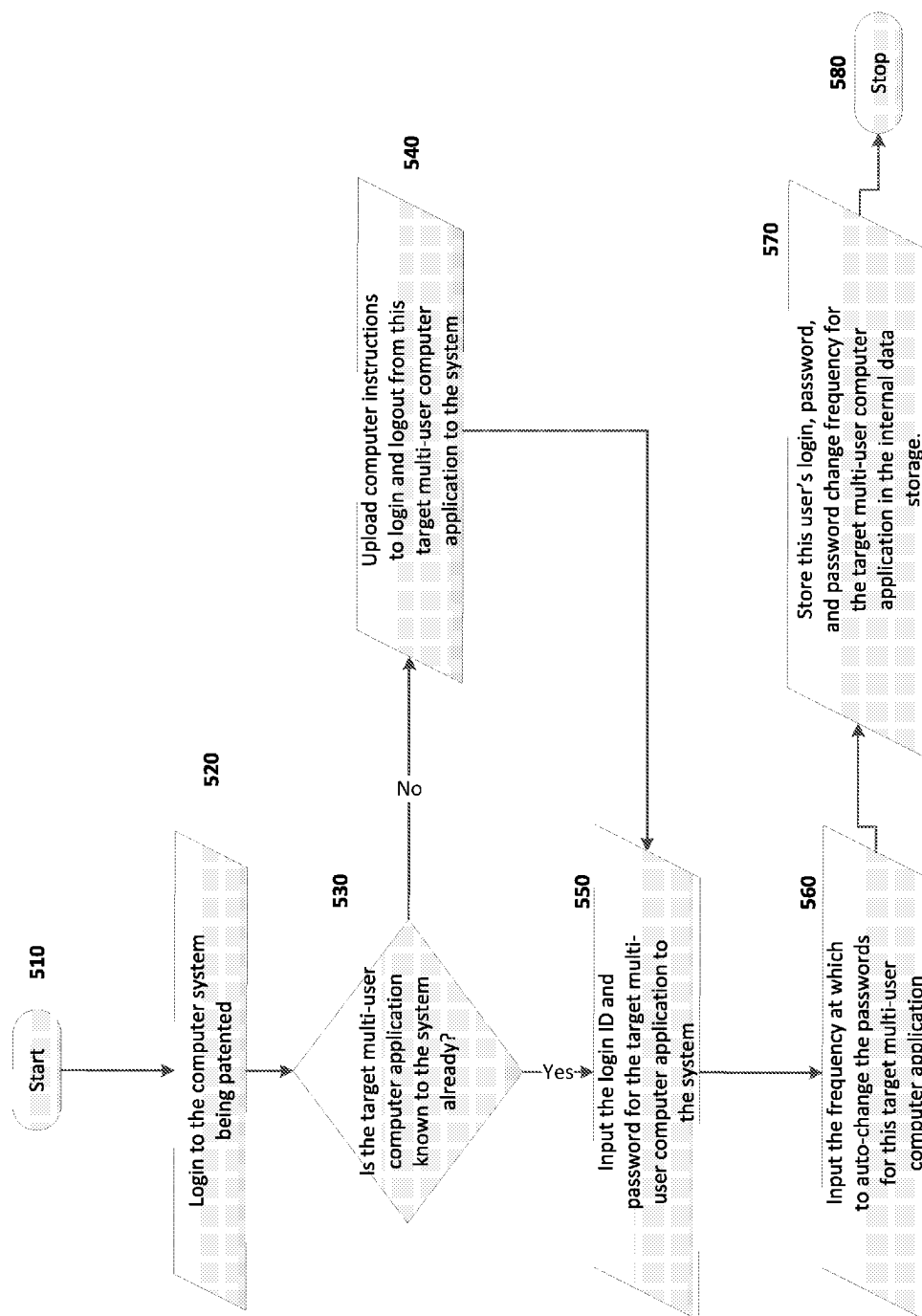
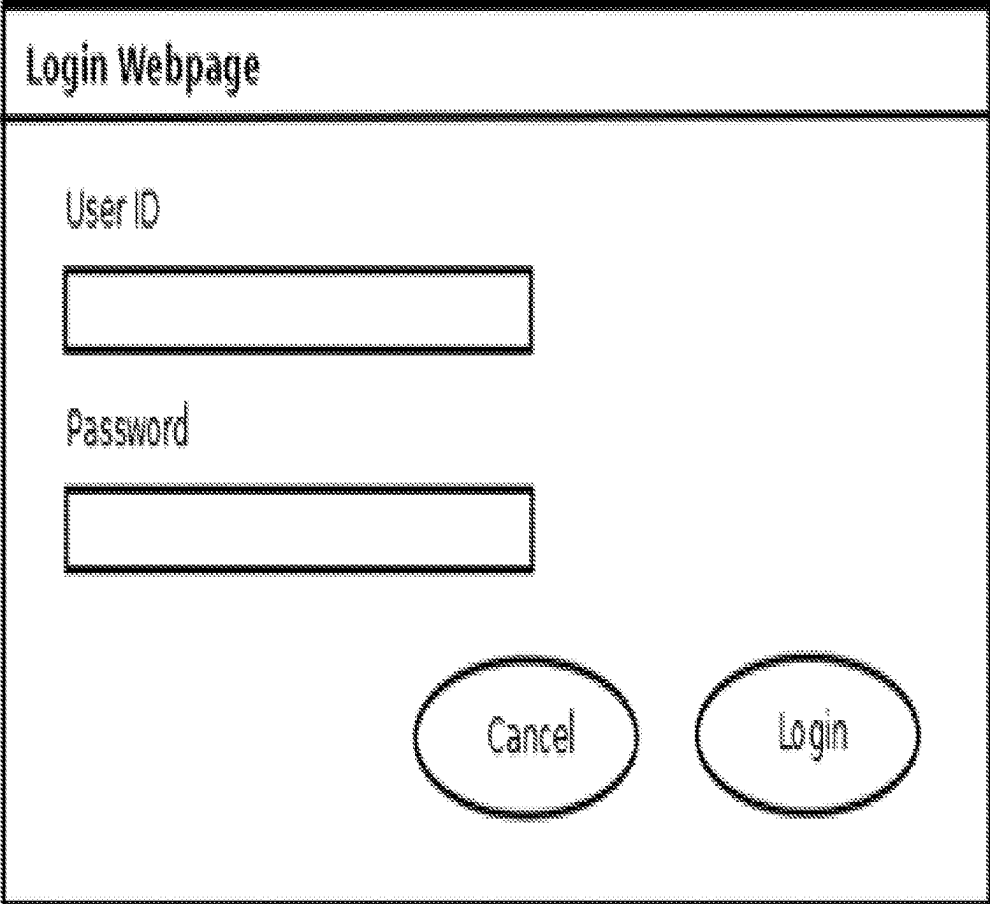


FIG. 5



The diagram illustrates a login webpage interface. It features a title bar at the top labeled "Login Webpage". Below the title bar, there are two input fields: the first is labeled "User ID" and the second is labeled "Password". At the bottom of the interface, there are two buttons: "Cancel" and "Login".

Login Webpage

User ID

Password

Cancel Login

FIG. 6

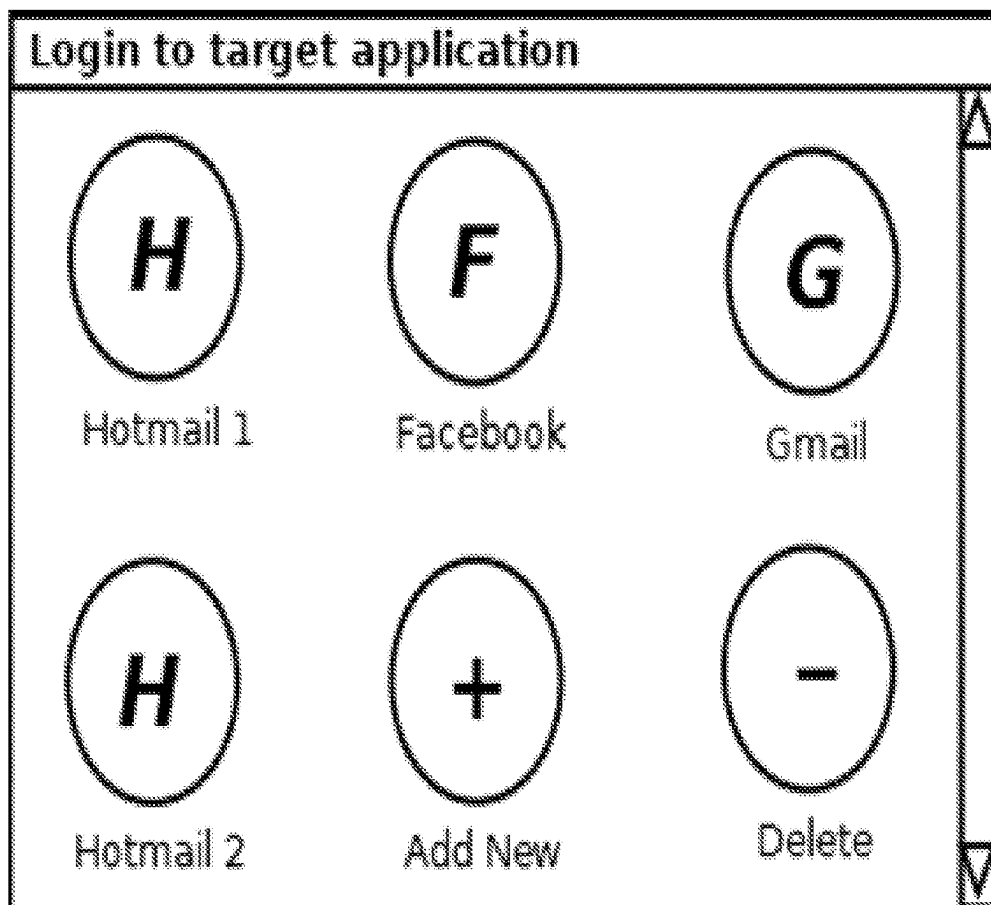


FIG. 7

Register new target application

Select Target Computer Application

H

Hotmail 1

F

Facebook

G

Gmail

Enter login ID for target application

Enter initial password for target application

Cancel

OK

FIG. 8

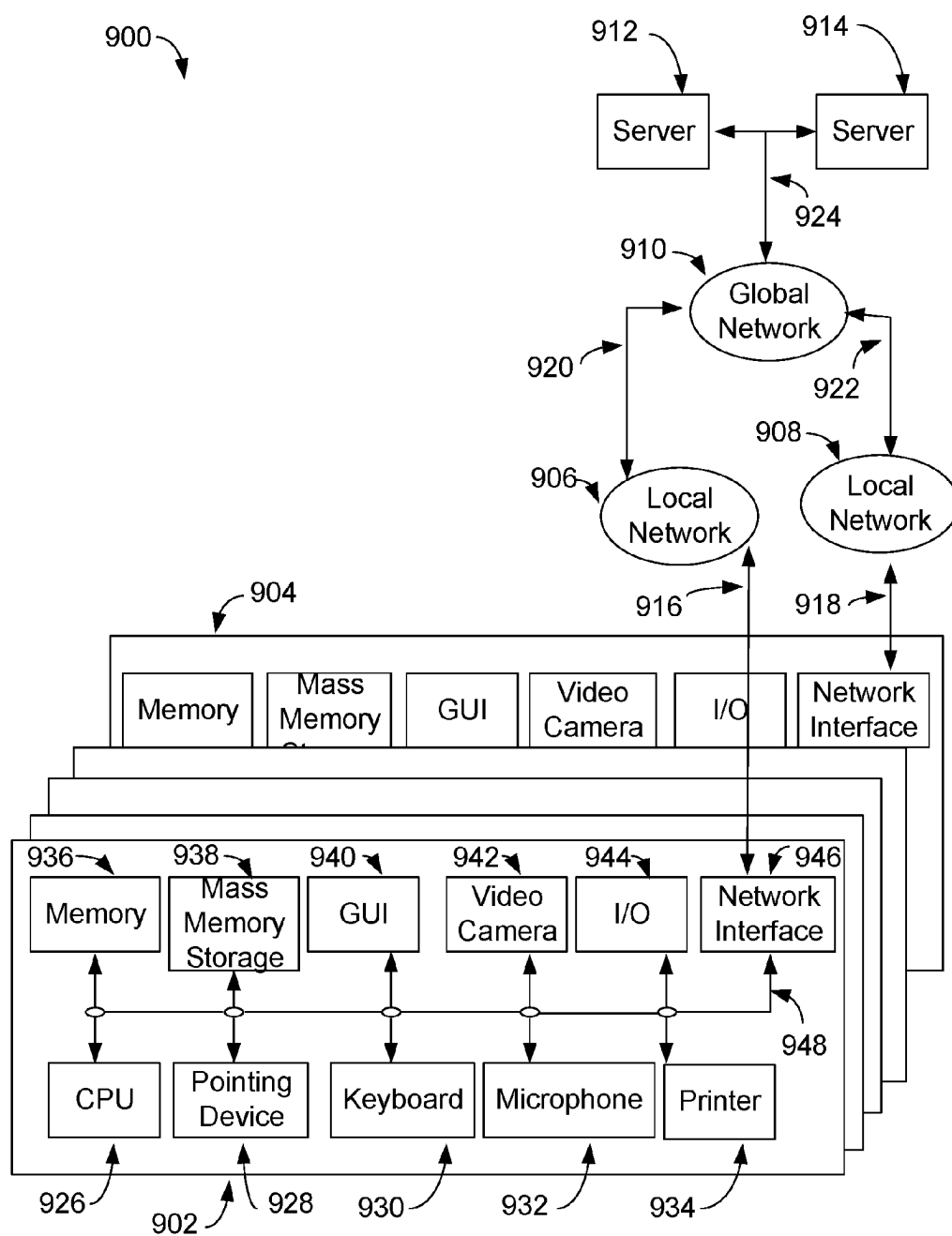


FIG. 9

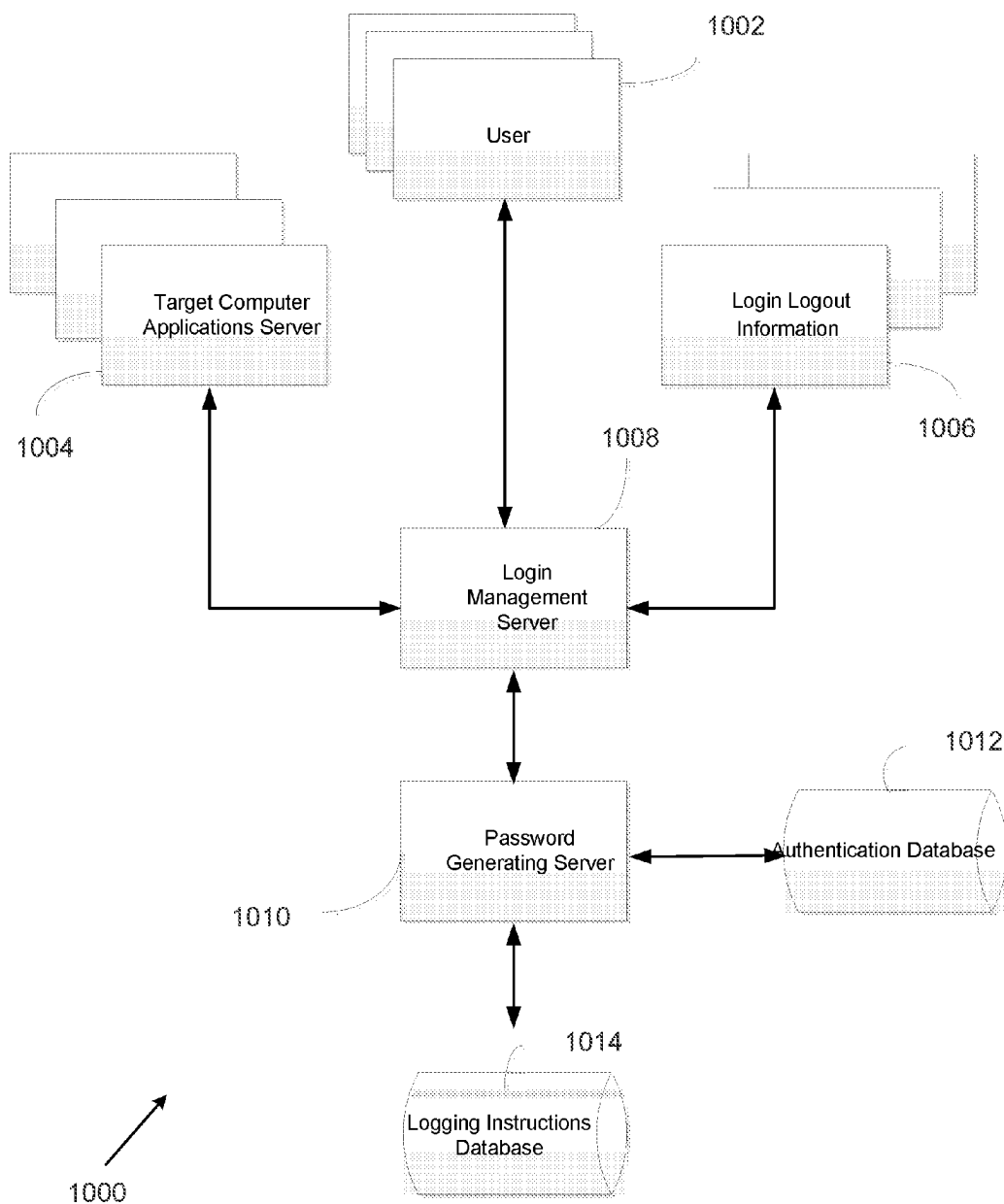


FIG. 10

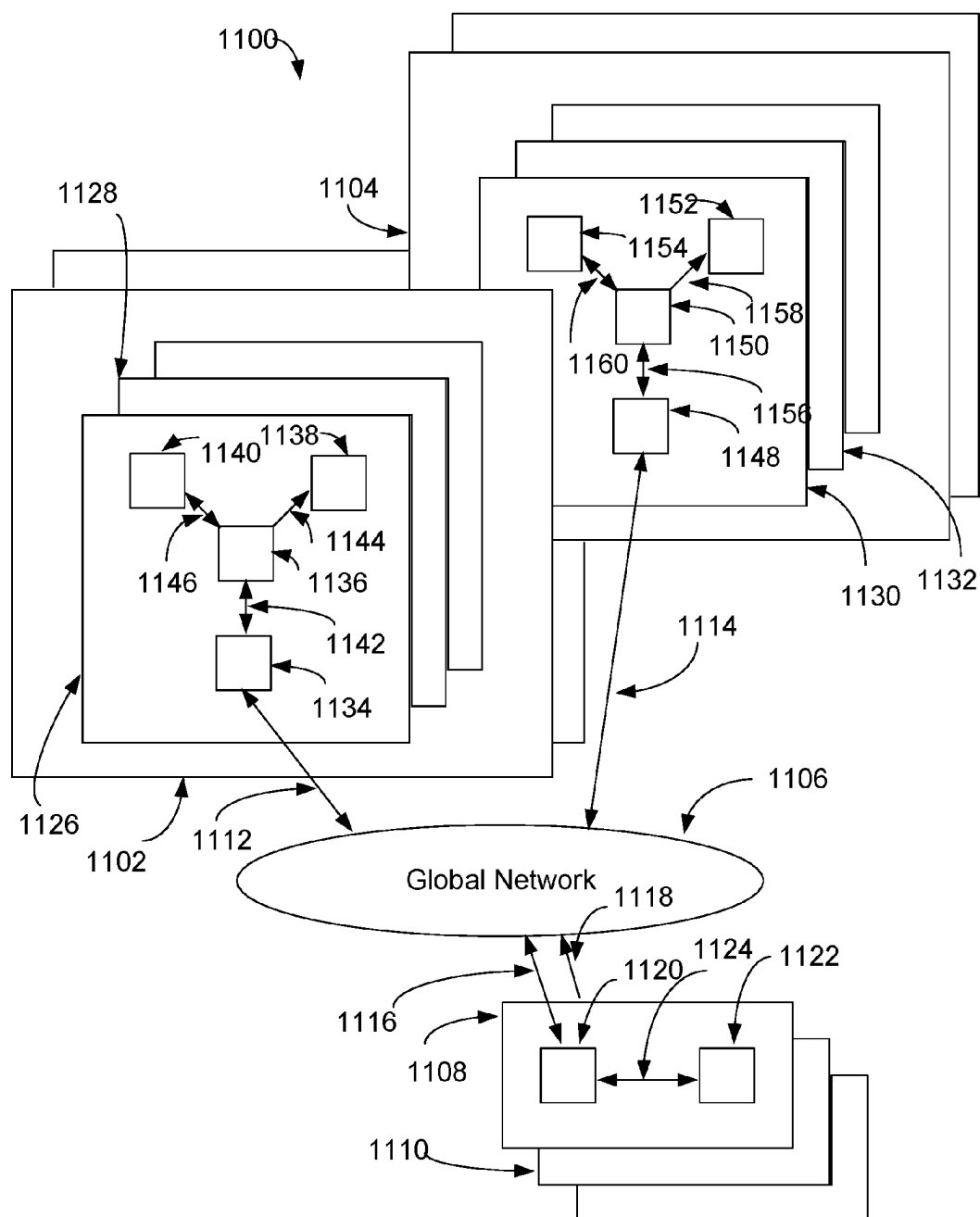


FIG. 11

SECURED APPLICATION ACCESS SYSTEM AND METHOD WITH FREQUENTLY CHANGING PASSWORDS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present Utility patent application claims priority benefit of the U.S. provisional application for patent Ser. No. 62/181,921 entitled "METHOD AND APPARATUS FOR SECURING AUTHENTICATION PASSWORD BY AUTOMATED FREQUENT MODIFICATIONS" filed 19 Jun. 2015 under 35 U.S.C. 119(e). The contents of this related provisional application are incorporated herein by reference for all purposes to the extent that such subject matter is not inconsistent herewith or limiting hereof.

RELATED CO-PENDING U.S. PATENT APPLICATIONS

[0002] Not applicable.

INCORPORATION BY REFERENCE OF SEQUENCE LISTING PROVIDED AS A TEXT FILE

[0003] Not applicable.

FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

[0004] Not applicable.

REFERENCE TO SEQUENCE LISTING, A TABLE, OR A COMPUTER LISTING APPENDIX

[0005] Not applicable.

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BACKGROUND OF THE RELEVANT PRIOR ART

[0007] One or more embodiments of the invention generally relate to computer programs for authenticating, authorizing, and identifying the users of a multi-user computer application and securing their passwords by changing them. More particularly, certain embodiments of the invention relate to continuously or frequently changing users' passwords for external multi-user computer applications, while still allowing the users to transparently access these external multi-user applications.

BACKGROUND OF THE INVENTION

[0008] The following background information may present examples of specific aspects of the prior art (e.g., without limitation, approaches, facts, or common wisdom) that, while expected to be helpful to further educate the reader as to additional aspects of the prior art, is not to be construed

as limiting the present invention, or any embodiments thereof, to anything stated or implied therein or inferred thereupon.

[0009] Some existing security applications use a single sign on software to enable users to access multiple multi-user applications by logging into a single application which then transfers the user session or authenticates them to another application. However existing security applications don't automatically change user's passwords. Even when the password policy is defined, at most the current security applications prompt the user to manually set or select a new password, when it is about to expire.

[0010] In this document, a multi-user computer application is defined as a computer application or system which allows multiple users to log into it and utilize its functionality. Common everyday examples of such applications are Google Email, Facebook etc.

[0011] The following is an example of a specific aspect in the prior art that, while expected to be helpful to further educate the reader as to additional aspects of the prior art, is not to be construed as limiting the present invention, or any embodiments thereof, to anything stated or implied therein or inferred thereupon. By way of educational background, another aspect of the prior art generally useful to be aware of is that a password may be generated, but only when the user initiates the password change.

[0012] In view of the foregoing, it is clear that these traditional techniques are not perfect and leave room for more optimal approaches for managing user security for multi-user computer applications.

BRIEF DESCRIPTION OF THE DRAWINGS

[0013] The present invention is illustrated by way of example, and not by way of limitation, in the figures of the accompanying drawings and in which like reference numerals refer to similar elements and in which:

[0014] FIG. 1 is an illustration of an exemplary computer system, in accordance with an embodiment of the present invention;

[0015] FIG. 2 is a flowchart of an exemplary method to frequently change a user's password for target computer applications, in accordance with an embodiment of the present invention;

[0016] FIG. 3 is a flowchart of an exemplary method for changing a user's password for all users and all registered target computer applications, in accordance with an embodiment of the present invention;

[0017] FIG. 4 is a flowchart of an exemplary method to allow a user to still login to their target multi-user computer application, in accordance with an embodiment of the present invention, without knowing their current password for it;

[0018] FIG. 5 is a flowchart of an exemplary method to allow a user to identify a target computer application, optionally specify the login, logout, and password change instructions, as well as the initial login password, in accordance with an embodiment of the present invention;

[0019] FIG. 6 is an illustration of an exemplary user interface of an Internet website allowing users to login to a computer system, in accordance with an embodiment of the present invention;

[0020] FIG. 7 is an illustration of an exemplary user interface of an Internet website allowing users to access their target computer applications with changed passwords, in accordance with an embodiment of the present invention;

[0021] FIG. 8 is an illustration of an exemplary user interface of an Internet website allowing users to register new target computer applications, in accordance with an embodiment of the present invention;

[0022] FIG. 9 is a block diagram depicting an exemplary client/server system which may be used by an exemplary web-enabled/networked embodiment of the present invention;

[0023] FIG. 10 illustrates a block diagram depicting an exemplary regionalized client/server communication system supporting authentication phrase verification and creation, in accordance with an embodiment of the present invention; and

[0024] FIG. 11 illustrates a block diagram depicting a conventional client/server communication system.

[0025] Unless otherwise indicated illustrations in the figures are not necessarily drawn to scale.

DETAILED DESCRIPTION OF SOME EMBODIMENTS

[0026] The present invention is best understood by reference to the detailed figures and description set forth herein.

[0027] Embodiments of the invention are discussed below with reference to the Figures. However, those skilled in the art will readily appreciate that the detailed description given herein with respect to these figures is for explanatory purposes as the invention extends beyond these limited embodiments. For example, it should be appreciated that those skilled in the art will, in light of the teachings of the present invention, recognize a multiplicity of alternate and suitable approaches, depending upon the needs of the particular application, to implement the functionality of any given detail described herein, beyond the particular implementation choices in the following embodiments described and shown. That is, there are modifications and variations of the invention that are too numerous to be listed but that all fit within the scope of the invention. Also, singular words should be read as plural and vice versa and masculine as feminine and vice versa, where appropriate, and alternative embodiments do not necessarily imply that the two are mutually exclusive.

[0028] It is to be further understood that the present invention is not limited to the particular methodology, compounds, materials, manufacturing techniques, uses, and applications, described herein, as these may vary. It is also to be understood that the terminology used herein is used for the purpose of describing particular embodiments only, and is not intended to limit the scope of the present invention. It must be noted that as used herein and in the appended claims, the singular forms “a,” “an,” and “the” include the plural reference unless the context clearly dictates otherwise. Thus, for example, a reference to “an element” is a reference to one or more elements and includes equivalents thereof known to those skilled in the art. Similarly, for another example, a reference to “a step” or “a means” is a reference to one or more steps or means and may include sub-steps and subservient means. All conjunctions used are to be understood in the most inclusive sense possible. Thus, the word “or” should be understood as having the definition of a logical “or” rather than that of a logical “exclusive or” unless the context clearly necessitates otherwise. Structures described herein are to be understood also to refer to functional equivalents of such structures. Language that may

be construed to express approximation should be so understood unless the context clearly dictates otherwise.

[0029] All words of approximation as used in the present disclosure and claims should be construed to mean “approximate,” rather than “perfect,” and may accordingly be employed as a meaningful modifier to any other word, specified parameter, quantity, quality, or concept. Words of approximation, include, yet are not limited to terms such as “substantial,” “nearly,” “almost,” “about,” “generally,” “largely,” “essentially,” “closely approximate”, etc.

[0030] As will be established in some detail below, it is well settled law, as early as 1939, that words of approximation are not indefinite in the claims even when such limits are not defined or specified in the specification.

[0031] For example, see *Ex parte Mallory*, 52 USPQ 297, 297 (Pat. Off. Bd. App. 1941) where the court said “The examiner has held that most of the claims are inaccurate because apparently the laminar film will not be entirely eliminated. The claims specify that the film is “substantially” eliminated and for the intended purpose, it is believed that the slight portion of the film which may remain is negligible. We are of the view, therefore, that the claims may be regarded as sufficiently accurate.”

[0032] Note that claims need only “reasonably apprise those skilled in the art” as to their scope to satisfy the definiteness requirement. See *Energy Absorption Sys., Inc. v. Roadway Safety Servs., Inc.*, Civ. App. 96-1264, slip op. at 10 (Fed. Cir. Jul. 3, 1997) (unpublished) *Hybridtech v. Monoclonal Antibodies, Inc.*, 802 F.2d 1367, 1385, 231 USPQ 81, 94 (Fed. Cir. 1986), cert. denied, 480 U.S. 947 (1987). In addition, the use of modifiers in the claim, like “generally” and “substantial,” does not by itself render the claims indefinite. See *Seattle Box Co. v. Industrial Crating & Packing, Inc.*, 731 F.2d 818, 828-29, 221 USPQ 568, 575-76 (Fed. Cir. 1984).

[0033] Moreover, the ordinary and customary meaning of terms like “substantially” includes “reasonably close to: nearly, almost, about”, connoting a term of approximation. See *In re Frye*, Appeal No. 2009-006013, 94 USPQ2d 1072, 1077, 2010 WL 889747 (B.P.A.I. 2010) Depending on its usage, the word “substantially” can denote either language of approximation or language of magnitude. *Deering Precision Instruments, L.L.C. v. Vector Distribution Sys., Inc.*, 347 F.3d 1314, 1323 (Fed. Cir. 2003) (recognizing the “dual ordinary meaning of th[e] term [“substantially”] as connoting a term of approximation or a term of magnitude”). Here, when referring to the “substantially halfway” limitation, the Specification uses the word “approximately” as a substitute for the word “substantially” (Fact 4). (Fact 4). The ordinary meaning of “substantially halfway” is thus reasonably close to or nearly at the midpoint between the forwardmost point of the upper or outsole and the rearwardmost point of the upper or outsole.

[0034] Similarly, the term ‘substantially’ is well recognized in case law to have the dual ordinary meaning of connoting a term of approximation or a term of magnitude. See *Dana Corp. v. American Axle & Manufacturing, Inc.*, Civ. App. 04-1116, 2004 U.S. App. LEXIS 18265, *13-14 (Fed. Cir. Aug. 27, 2004) (unpublished). The term “substantially” is commonly used by claim drafters to indicate approximation. See *Cordis Corp. v. Medtronic AVE Inc.*, 339 F.3d 1352, 1360 (Fed. Cir. 2003) (“The patents do not set out any numerical standard by which to determine whether the thickness of the wall surface is ‘substantially uniform.’ The

term ‘substantially,’ as used in this context, denotes approximation. Thus, the walls must be of largely or approximately uniform thickness.”); see also *Deering Precision Instruments, LLC v. Vector Distribution Sys., Inc.*, 347 F.3d 1314, 1322 (Fed. Cir. 2003); *Epcon Gas Sys., Inc. v. Bauer Compressors, Inc.*, 279 F.3d 1022, 1031 (Fed. Cir. 2002). We find that the term “substantially” was used in just such a manner in the claims of the patents-in-suit: “substantially uniform wall thickness” denotes a wall thickness with approximate uniformity.

[0035] It should also be noted that such words of approximation as contemplated in the foregoing clearly limits the scope of claims such as saying ‘generally parallel’ such that the adverb ‘generally’ does not broaden the meaning of parallel. Accordingly, it is well settled that such words of approximation as contemplated in the foregoing (e.g., like the phrase ‘generally parallel’) envisions some amount of deviation from perfection (e.g., not exactly parallel), and that such words of approximation as contemplated in the foregoing are descriptive terms commonly used in patent claims to avoid a strict numerical boundary to the specified parameter. To the extent that the plain language of the claims relying on such words of approximation as contemplated in the foregoing are clear and uncontradicted by anything in the written description herein or the figures thereof, it is improper to rely upon the present written description, the figures, or the prosecution history to add limitations to any of the claim of the present invention with respect to such words of approximation as contemplated in the foregoing. That is, under such circumstances, relying on the written description and prosecution history to reject the ordinary and customary meanings of the words themselves is impermissible. See, for example, *Liquid Dynamics Corp. v. Vaughan Co.*, 355 F.3d 1361, 69 USPQ2d 1595, 1600-01 (Fed. Cir. 2004). The plain language of phrase 2 requires a “substantial helical flow.” The term “substantial” is a meaningful modifier implying “approximate,” rather than “perfect.” In *Cordis Corp. v. Medtronic AVE, Inc.*, 339 F.3d 1352, 1361 (Fed. Cir. 2003), the district court imposed a precise numeric constraint on the term “substantially uniform thickness.” We noted that the proper interpretation of this term was “of largely or approximately uniform thickness” unless something in the prosecution history imposed the “clear and unmistakable disclaimer” needed for narrowing beyond this simple-language interpretation. *Id.* In *Anchor Wall Systems v. Rockwood Retaining Walls, Inc.*, 340 F.3d 1298, 1311 (Fed. Cir. 2003)” *Id.* at 1311. Similarly, the plain language of claim 1 requires neither a perfectly helical flow nor a flow that returns precisely to the center after one rotation (a limitation that arises only as a logical consequence of requiring a perfectly helical flow).

[0036] The reader should appreciate that case law generally recognizes a dual ordinary meaning of such words of approximation, as contemplated in the foregoing, as connoting a term of approximation or a term of magnitude; e.g., see *Deering Precision Instruments, L.L.C. v. Vector Distrib. Sys., Inc.*, 347 F.3d 1314, 68 USPQ2d 1716, 1721 (Fed. Cir. 2003), cert. denied, 124 S. Ct. 1426 (2004) where the court was asked to construe the meaning of the term “substantially” in a patent claim. Also see *Epcon*, 279 F.3d at 1031 (“The phrase ‘substantially constant’ denotes language of approximation, while the phrase ‘substantially below’ signifies language of magnitude, i.e., not insubstantial.”). Also, see, e.g., *Epcon Gas Sys., Inc. v. Bauer Compressors, Inc.*,

279 F.3d 1022 (Fed. Cir. 2002) (construing the terms “substantially constant” and “substantially below”); *Zodiac Pool Care, Inc. v. Hoffinger Indus., Inc.*, 206 F.3d 1408 (Fed. Cir. 2000) (construing the term “substantially inward”); *York Prods., Inc. v. Cent. Tractor Farm & Family Ctr.*, 99 F.3d 1568 (Fed. Cir. 1996) (construing the term “substantially the entire height thereof”); *Tex. Instruments Inc. v. Cypress Semiconductor Corp.*, 90 F.3d 1558 (Fed. Cir. 1996) (construing the term “substantially in the common plane”). In conducting their analysis, the court instructed to begin with the ordinary meaning of the claim terms to one of ordinary skill in the art. *Prima Tek*, 318 F.3d at 1148. Reference to dictionaries and our cases indicates that the term “substantially” has numerous ordinary meanings. As the district court stated, “substantially” can mean “significantly” or “considerably.” The term “substantially” can also mean “largely” or “essentially.” Webster’s New 20th Century Dictionary 1817 (1983).

[0037] Words of approximation, as contemplated in the foregoing, may also be used in phrases establishing approximate ranges or limits, where the end points are inclusive and approximate, not perfect; e.g., see *AK Steel Corp. v. Sollac*, 344 F.3d 1234, 68 USPQ2d 1280, 1285 (Fed. Cir. 2003) where it where the court said [W]e conclude that the ordinary meaning of the phrase “up to about 10%” includes the “about 10%” endpoint. As pointed out by AK Steel, when an object of the preposition “up to” is nonnumeric, the most natural meaning is to exclude the object (e.g., painting the wall up to the door). On the other hand, as pointed out by Sollac, when the object is a numerical limit, the normal meaning is to include that upper numerical limit (e.g., counting up to ten, seating capacity for up to seven passengers). Because we have here a numerical limit—“about 10%”—the ordinary meaning is that that endpoint is included.

[0038] In the present specification and claims, a goal of employment of such words of approximation, as contemplated in the foregoing, is to avoid a strict numerical boundary to the modified specified parameter, as sanctioned by *Pall Corp. v. Micron Separations, Inc.*, 66 F.3d 1211, 1217, 36 USPQ2d 1225, 1229 (Fed. Cir. 1995) where it states “It is well established that when the term “substantially” serves reasonably to describe the subject matter so that its scope would be understood by persons in the field of the invention, and to distinguish the claimed subject matter from the prior art, it is not indefinite.” Likewise see *Verve LLC v. Crane Cams Inc.*, 311 F.3d 1116, 65 USPQ2d 1051, 1054 (Fed. Cir. 2002). Expressions such as “substantially” are used in patent documents when warranted by the nature of the invention, in order to accommodate the minor variations that may be appropriate to secure the invention. Such usage may well satisfy the charge to “particularly point out and distinctly claim” the invention, 35 U.S.C. §112, and indeed may be necessary in order to provide the inventor with the benefit of his invention. In *Andrew Corp. v. Gabriel Elecs. Inc.*, 847 F.2d 819, 821-22, 6 USPQ2d 2010, 2013 (Fed. Cir. 1988) the court explained that usages such as “substantially equal” and “closely approximate” may serve to describe the invention with precision appropriate to the technology and without intruding on the prior art. The court again explained in *Ecolab Inc. v. Envirochem, Inc.*, 264 F.3d 1358, 1367, 60 USPQ2d 1173, 1179 (Fed. Cir. 2001) that “like the term ‘about,’ the term ‘substantially’ is a descriptive term commonly used in patent claims to ‘avoid a strict

numerical boundary to the specified parameter, see *Ecolab Inc. v. Envirochem Inc.*, 264 F.3d 1358, 60 USPQ2d 1173, 1179 (Fed. Cir. 2001) where the court found that the use of the term “substantially” to modify the term “uniform” does not render this phrase so unclear such that there is no means by which to ascertain the claim scope.

[0039] Similarly, other courts have noted that like the term “about,” the term “substantially” is a descriptive term commonly used in patent claims to “avoid a strict numerical boundary to the specified parameter.”; e.g., see *Pall Corp. v. Micron Seps.*, 66 F.3d 1211, 1217, 36 USPQ2d 1225, 1229 (Fed. Cir. 1995); see, e.g., *Andrew Corp. v. Gabriel Elecs. Inc.*, 847 F.2d 819, 821-22, 6 USPQ2d 2010, 2013 (Fed. Cir. 1988) (noting that terms such as “approach each other,” “close to,” “substantially equal,” and “closely approximate” are ubiquitously used in patent claims and that such usages, when serving reasonably to describe the claimed subject matter to those of skill in the field of the invention, and to distinguish the claimed subject matter from the prior art, have been accepted in patent examination and upheld by the courts). In this case, “substantially” avoids the strict 100% nonuniformity boundary.

[0040] Indeed, the foregoing sanctioning of such words of approximation, as contemplated in the foregoing, has been established as early as 1939, see *Ex parte Mallory*, 52 USPQ 297, 297 (Pat. Off. Bd. App. 1941) where, for example, the court said “the claims specify that the film is “substantially” eliminated and for the intended purpose, it is believed that the slight portion of the film which may remain is negligible. We are of the view, therefore, that the claims may be regarded as sufficiently accurate.” Similarly, In *re Hutchison*, 104 F.2d 829, 42 USPQ 90, 93 (C.C.P.A. 1939) the court said “It is realized that “substantial distance” is a relative and somewhat indefinite term, or phrase, but terms and phrases of this character are not uncommon in patents in cases where, according to the art involved, the meaning can be determined with reasonable clearness.”

[0041] Hence, for at least the foregoing reason, Applicants submit that it is improper for any examiner to hold as indefinite any claims of the present patent that employ any words of approximation.

[0042] Unless defined otherwise, all technical and scientific terms used herein have the same meanings as commonly understood by one of ordinary skill in the art to which this invention belongs. Preferred methods, techniques, devices, and materials are described, although any methods, techniques, devices, or materials similar or equivalent to those described herein may be used in the practice or testing of the present invention. Structures described herein are to be understood also to refer to functional equivalents of such structures. The present invention will be described in detail below with reference to embodiments thereof as illustrated in the accompanying drawings.

[0043] References to a “device,” an “apparatus,” a “system,” etc., in the preamble of a claim should be construed broadly to mean “any structure meeting the claim terms” exempt for any specific structure(s)/type(s) that has/(have) been explicitly disavowed or excluded or admitted/implicit as prior art in the present specification or incapable of enabling an object/aspect/goal of the invention. Furthermore, where the present specification discloses an object, aspect, function, goal, result, or advantage of the invention that a specific prior art structure and/or method step is similarly capable of performing yet in a very different way,

the present invention disclosure is intended to and shall also implicitly include and cover additional corresponding alternative embodiments that are otherwise identical to that explicitly disclosed except that they exclude such prior art structure(s)/step(s), and shall accordingly be deemed as providing sufficient disclosure to support a corresponding negative limitation in a claim claiming such alternative embodiment(s), which exclude such very different prior art structure(s)/step(s) way(s).

[0044] From reading the present disclosure, other variations and modifications will be apparent to persons skilled in the art. Such variations and modifications may involve equivalent and other features which are already known in the art, and which may be used instead of or in addition to features already described herein.

[0045] Although Claims have been formulated in this application to particular combinations of features, it should be understood that the scope of the disclosure of the present invention also includes any novel feature or any novel combination of features disclosed herein either explicitly or implicitly or any generalization thereof, whether or not it relates to the same invention as presently claimed in any Claim and whether or not it mitigates any or all of the same technical problems as does the present invention.

[0046] Features which are described in the context of separate embodiments may also be provided in combination in a single embodiment. Conversely, various features which are, for brevity, described in the context of a single embodiment, may also be provided separately or in any suitable subcombination. The Applicants hereby give notice that new Claims may be formulated to such features and/or combinations of such features during the prosecution of the present application or of any further application derived therefrom.

[0047] References to “one embodiment,” “an embodiment,” “example embodiment,” “various embodiments,” “some embodiments,” “embodiments of the invention,” etc., may indicate that the embodiment(s) of the invention so described may include a particular feature, structure, or characteristic, but not every possible embodiment of the invention necessarily includes the particular feature, structure, or characteristic. Further, repeated use of the phrase “in one embodiment,” or “in an exemplary embodiment,” “an embodiment,” do not necessarily refer to the same embodiment, although they may. Moreover, any use of phrases like “embodiments” in connection with “the invention” are never meant to characterize that all embodiments of the invention must include the particular feature, structure, or characteristic, and should instead be understood to mean “at least some embodiments of the invention” includes the stated particular feature, structure, or characteristic.

[0048] References to “user”, or any similar term, as used herein, may mean a human or non-human user thereof. Moreover, “user”, or any similar term, as used herein, unless expressly stipulated otherwise, is contemplated to mean users at any stage of the usage process, to include, without limitation, direct user(s), intermediate user(s), indirect user(s), and end user(s). The meaning of “user”, or any similar term, as used herein, should not be otherwise inferred or induced by any pattern(s) of description, embodiments, examples, or referenced prior-art that may (or may not) be provided in the present patent.

[0049] References to “end user”, or any similar term, as used herein, is generally intended to mean late stage user(s) as opposed to early stage user(s). Hence, it is contemplated

that there may be a multiplicity of different types of “end user” near the end stage of the usage process. Where applicable, especially with respect to distribution channels of embodiments of the invention comprising consumed retail products/services thereof (as opposed to sellers/vendors or Original Equipment Manufacturers), examples of an “end user” may include, without limitation, a “consumer”, “buyer”, “customer”, “purchaser”, “shopper”, “enjoyer”, “viewer”, or individual person or non-human thing benefiting in any way, directly or indirectly, from use of, or interaction, with some aspect of the present invention.

[0050] In some situations, some embodiments of the present invention may provide beneficial usage to more than one stage or type of usage in the foregoing usage process. In such cases where multiple embodiments targeting various stages of the usage process are described, references to “end user”, or any similar term, as used therein, are generally intended to not include the user that is the furthest removed, in the foregoing usage process, from the final user therein of an embodiment of the present invention.

[0051] Where applicable, especially with respect to retail distribution channels of embodiments of the invention, intermediate user(s) may include, without limitation, any individual person or non-human thing benefiting in any way, directly or indirectly, from use of, or interaction with, some aspect of the present invention with respect to selling, vending, Original Equipment Manufacturing, marketing, merchandising, distributing, service providing, and the like thereof.

[0052] References to “person”, “individual”, “human”, “a party”, “animal”, “creature”, or any similar term, as used herein, even if the context or particular embodiment implies living user, maker, or participant, it should be understood that such characterizations are sole by way of example, and not limitation, in that it is contemplated that any such usage, making, or participation by a living entity in connection with making, using, and/or participating, in any way, with embodiments of the present invention may be substituted by such similar performed by a suitably configured non-living entity, to include, without limitation, automated machines, robots, humanoids, computational systems, information processing systems, artificially intelligent systems, and the like. It is further contemplated that those skilled in the art will readily recognize the practical situations where such living makers, users, and/or participants with embodiments of the present invention may be in whole, or in part, replaced with such non-living makers, users, and/or participants with embodiments of the present invention. Likewise, when those skilled in the art identify such practical situations where such living makers, users, and/or participants with embodiments of the present invention may be in whole, or in part, replaced with such non-living makers, it will be readily apparent in light of the teachings of the present invention how to adapt the described embodiments to be suitable for such non-living makers, users, and/or participants with embodiments of the present invention. Thus, the invention is thus to also cover all such modifications, equivalents, and alternatives falling within the spirit and scope of such adaptations and modifications, at least in part, for such non-living entities.

[0053] Headings provided herein are for convenience and are not to be taken as limiting the disclosure in any way.

[0054] The enumerated listing of items does not imply that any or all of the items are mutually exclusive, unless expressly specified otherwise.

[0055] It is understood that the use of specific component, device and/or parameter names are for example only and not meant to imply any limitations on the invention. The invention may thus be implemented with different nomenclature/terminology utilized to describe the mechanisms/units/structures/components/devices/parameters herein, without limitation. Each term utilized herein is to be given its broadest interpretation given the context in which that term is utilized.

TERMINOLOGY

[0056] The following paragraphs provide definitions and/or context for terms found in this disclosure (including the appended claims):

[0057] “Comprising.” This term is open-ended. As used in the appended claims, this term does not foreclose additional structure or steps. Consider a claim that recites: “A memory controller comprising a system cache” Such a claim does not foreclose the memory controller from including additional components (e.g., a memory channel unit, a switch).

[0058] “Configured To.” Various units, circuits, or other components may be described or claimed as “configured to” perform a task or tasks. In such contexts, “configured to” or “operable for” is used to connote structure by indicating that the mechanisms/units/circuits/components include structure (e.g., circuitry and/or mechanisms) that performs the task or tasks during operation. As such, the mechanisms/unit/circuit/component can be said to be configured to (or be operable) for perform(ing) the task even when the specified mechanisms/unit/circuit/component is not currently operational (e.g., is not on). The mechanisms/units/circuits/components used with the “configured to” or “operable for” language include hardware—for example, mechanisms, structures, electronics, circuits, memory storing program instructions executable to implement the operation, etc. Reciting that a mechanism/unit/circuit/component is “configured to” or “operable for” perform(ing) one or more tasks is expressly intended not to invoke 35 U.S.C. sectn. 112, sixth paragraph, for that mechanism/unit/circuit/component. “Configured to” may also include adapting a manufacturing process to fabricate devices or components that are adapted to implement or perform one or more tasks.

[0059] “Based On.” As used herein, this term is used to describe one or more factors that affect a determination. This term does not foreclose additional factors that may affect a determination. That is, a determination may be solely based on those factors or based, at least in part, on those factors. Consider the phrase “determine A based on B.” While B may be a factor that affects the determination of A, such a phrase does not foreclose the determination of A from also being based on C. In other instances, A may be determined based solely on B.

[0060] The terms “a”, “an” and “the” mean “one or more”, unless expressly specified otherwise.

[0061] Unless otherwise indicated, all numbers expressing conditions, concentrations, dimensions, and so forth used in the specification and claims are to be understood as being modified in all instances by the term “about.” Accordingly, unless indicated to the contrary, the numerical parameters set

forth in the following specification and attached claims are approximations that may vary depending at least upon a specific analytical technique.

[0062] The term “comprising,” which is synonymous with “including,” “containing,” or “characterized by” is inclusive or open-ended and does not exclude additional, unrecited elements or method steps. “Comprising” is a term of art used in claim language which means that the named claim elements are essential, but other claim elements may be added and still form a construct within the scope of the claim.

[0063] As used herein, the phrase “consisting of” excludes any element, step, or ingredient not specified in the claim. When the phrase “consists of” (or variations thereof) appears in a clause of the body of a claim, rather than immediately following the preamble, it limits only the element set forth in that clause; other elements are not excluded from the claim as a whole. As used herein, the phrase “consisting essentially of” and “consisting of” limits the scope of a claim to the specified elements or method steps, plus those that do not materially affect the basis and novel characteristic(s) of the claimed subject matter (see *Norian Corp. v Stryker Corp.*, 363 F.3d 1321, 1331-32, 70 USPQ2d 1508, Fed. Cir. 2004). Moreover, for any claim of the present invention which claims an embodiment “consisting essentially of” or “consisting of” a certain set of elements of any herein described embodiment it shall be understood as obvious by those skilled in the art that the present invention also covers all possible varying scope variants of any described embodiment(s) that are each exclusively (i.e., “consisting essentially of”) functional subsets or functional combination thereof such that each of these plurality of exclusive varying scope variants each consists essentially of any functional subset(s) and/or functional combination(s) of any set of elements of any described embodiment(s) to the exclusion of any others not set forth therein. That is, it is contemplated that it will be obvious to those skilled how to create a multiplicity of alternate embodiments of the present invention that simply consisting essentially of a certain functional combination of elements of any described embodiment(s) to the exclusion of any others not set forth therein, and the invention thus covers all such exclusive embodiments as if they were each described herein.

[0064] With respect to the terms “comprising,” “consisting of,” and “consisting essentially of” where one of these three terms is used herein, the presently disclosed and claimed subject matter may include the use of either of the other two terms. Thus in some embodiments not otherwise explicitly recited, any instance of “comprising” may be replaced by “consisting of” or, alternatively, by “consisting essentially of,” and thus, for the purposes of claim support and construction for “consisting of” format claims, such replacements operate to create yet other alternative embodiments “consisting essentially of” only the elements recited in the original “comprising” embodiment to the exclusion of all other elements.

[0065] Devices or system modules that are in at least general communication with each other need not be in continuous communication with each other, unless expressly specified otherwise. In addition, devices or system modules that are in at least general communication with each other may communicate directly or indirectly through one or more intermediaries.

[0066] A description of an embodiment with several components in communication with each other does not imply that all such components are required. On the contrary a variety of optional components are described to illustrate the wide variety of possible embodiments of the present invention.

[0067] As is well known to those skilled in the art many careful considerations and compromises typically must be made when designing for the optimal manufacture of a commercial implementation any system, and in particular, the embodiments of the present invention. A commercial implementation in accordance with the spirit and teachings of the present invention may be configured according to the needs of the particular application, whereby any aspect(s), feature(s), function(s), result(s), component(s), approach(es), or step(s) of the teachings related to any described embodiment of the present invention may be suitably omitted, included, adapted, mixed and matched, or improved and/or optimized by those skilled in the art, using their average skills and known techniques, to achieve the desired implementation that addresses the needs of the particular application.

[0068] In the following description and claims, the terms “coupled” and “connected,” along with their derivatives, may be used. It should be understood that these terms are not intended as synonyms for each other. Rather, in particular embodiments, “connected” may be used to indicate that two or more elements are in direct physical or electrical contact with each other. “Coupled” may mean that two or more elements are in direct physical or electrical contact. However, “coupled” may also mean that two or more elements are not in direct contact with each other, but yet still cooperate or interact with each other.

[0069] A “computer” may refer to one or more apparatus and/or one or more systems that are capable of accepting a structured input, processing the structured input according to prescribed rules, and producing results of the processing as output. Examples of a computer may include: a computer; a stationary and/or portable computer; a computer having a single processor, multiple processors, or multi-core processors, which may operate in parallel and/or not in parallel; a general purpose computer; a supercomputer; a mainframe; a super mini-computer; a mini-computer; a workstation; a micro-computer; a server; a client; an interactive television; a web appliance; a telecommunications device with internet access; a hybrid combination of a computer and an interactive television; a portable computer; a tablet personal computer (PC); a personal digital assistant (PDA); a portable telephone; application-specific hardware to emulate a computer and/or software, such as, for example, a digital signal processor (DSP), a field-programmable gate array (FPGA), an application specific integrated circuit (ASIC), an application specific instruction-set processor (ASIP), a chip, chips, a system on a chip, or a chip set; a data acquisition device; an optical computer; a quantum computer; a biological computer; and generally, an apparatus that may accept data, process data according to one or more stored software programs, generate results, and typically include input, output, storage, arithmetic, logic, and control units.

[0070] Those of skill in the art will appreciate that where appropriate, some embodiments of the disclosure may be practiced in network computing environments with many types of computer system configurations, including personal computers, hand-held devices, multi-processor systems,

microprocessor-based or programmable consumer electronics, network PCs, minicomputers, mainframe computers, and the like. Where appropriate, embodiments may also be practiced in distributed computing environments where tasks are performed by local and remote processing devices that are linked (either by hardwired links, wireless links, or by a combination thereof) through a communications network. In a distributed computing environment, program modules may be located in both local and remote memory storage devices.

[0071] “Software” may refer to prescribed rules to operate a computer. Examples of software may include: code segments in one or more computer-readable languages; graphical and/or textual instructions; applets; pre-compiled code; interpreted code; compiled code; and computer programs.

[0072] The example embodiments described herein can be implemented in an operating environment comprising computer-executable instructions (e.g., software) installed on a computer, in hardware, or in a combination of software and hardware. The computer-executable instructions can be written in a computer programming language or can be embodied in firmware logic. If written in a programming language conforming to a recognized standard, such instructions can be executed on a variety of hardware platforms and for interfaces to a variety of operating systems. Although not limited thereto, computer software program code for carrying out operations for aspects of the present invention can be written in any combination of one or more suitable programming languages, including an object oriented programming languages and/or conventional procedural programming languages, and/or programming languages such as, for example, Hypertext Markup Language (HTML), Dynamic HTML, Extensible Markup Language (XML), Extensible Stylesheet Language (XSL), Document Style Semantics and Specification Language (DSSSL), Cascading Style Sheets (CSS), Synchronized Multimedia Integration Language (SMIL), Wireless Markup Language (WML), Java™, Jini™, C, C++, Smalltalk, Perl, UNIX Shell, Visual Basic or Visual Basic Script, Virtual Reality Markup Language (VRML), ColdFusion™ or other compilers, assemblers, interpreters or other computer languages or platforms.

[0073] Computer program code for carrying out operations for aspects of the present invention may be written in any combination of one or more programming languages, including an object oriented programming language such as Java, Smalltalk, C++ or the like and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The program code may execute entirely on the user’s computer, partly on the user’s computer, as a stand-alone software package, partly on the user’s computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user’s computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

[0074] A network is a collection of links and nodes (e.g., multiple computers and/or other devices connected together) arranged so that information may be passed from one part of the network to another over multiple links and through various nodes. Examples of networks include the Internet, the public switched telephone network, the global Telex

network, computer networks (e.g., an intranet, an extranet, a local-area network, or a wide-area network), wired networks, and wireless networks.

[0075] The Internet is a worldwide network of computers and computer networks arranged to allow the easy and robust exchange of information between computer users. Hundreds of millions of people around the world have access to computers connected to the Internet via Internet Service Providers (ISPs). Content providers (e.g., website owners or operators) place multimedia information (e.g., text, graphics, audio, video, animation, and other forms of data) at specific locations on the Internet referred to as webpages. Websites comprise a collection of connected, or otherwise related, webpages. The combination of all the websites and their corresponding webpages on the Internet is generally known as the World Wide Web (WWW) or simply the Web.

[0076] Aspects of the present invention are described below with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according to embodiments of the invention. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0077] The flowchart and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods and computer program products according to various embodiments. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

[0078] These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the flowchart and/or block diagram block or blocks.

[0079] Further, although process steps, method steps, algorithms or the like may be described in a sequential order, such processes, methods and algorithms may be configured

to work in alternate orders. In other words, any sequence or order of steps that may be described does not necessarily indicate a requirement that the steps be performed in that order. The steps of processes described herein may be performed in any order practical. Further, some steps may be performed simultaneously.

[0080] It will be readily apparent that the various methods and algorithms described herein may be implemented by, e.g., appropriately programmed general purpose computers and computing devices. Typically, a processor (e.g., a micro-processor) will receive instructions from a memory or like device, and execute those instructions, thereby performing a process defined by those instructions. Further, programs that implement such methods and algorithms may be stored and transmitted using a variety of known media.

[0081] When a single device or article is described herein, it will be readily apparent that more than one device/article (whether or not they cooperate) may be used in place of a single device/article. Similarly, where more than one device or article is described herein (whether or not they cooperate), it will be readily apparent that a single device/article may be used in place of the more than one device or article.

[0082] The functionality and/or the features of a device may be alternatively embodied by one or more other devices which are not explicitly described as having such functionality/features. Thus, other embodiments of the present invention need not include the device itself.

[0083] The term “computer-readable medium” as used herein refers to any medium that participates in providing data (e.g., instructions) which may be read by a computer, a processor or a like device. Such a medium may take many forms, including but not limited to, non-volatile media, volatile media, and transmission media. Non-volatile media include, for example, optical or magnetic disks and other persistent memory. Volatile media include dynamic random access memory (DRAM), which typically constitutes the main memory. Transmission media include coaxial cables, copper wire and fiber optics, including the wires that comprise a system bus coupled to the processor. Transmission media may include or convey acoustic waves, light waves and electromagnetic emissions, such as those generated during radio frequency (RF) and infrared (IR) data communications. Common forms of computer-readable media include, for example, a floppy disk, a flexible disk, hard disk, magnetic tape, any other magnetic medium, a CD-ROM, DVD, any other optical medium, punch cards, paper tape, any other physical medium with patterns of holes, a RAM, a PROM, an EPROM, a FLASH-EEPROM, removable media, flash memory, a “memory stick”, any other memory chip or cartridge, a carrier wave as described hereinafter, or any other medium from which a computer can read.

[0084] Various forms of computer readable media may be involved in carrying sequences of instructions to a processor. For example, sequences of instruction (i) may be delivered from RAM to a processor, (ii) may be carried over a wireless transmission medium, and/or (iii) may be formatted according to numerous formats, standards or protocols, such as Bluetooth, TDMA, CDMA, 3G.

[0085] Where databases are described, it will be understood by one of ordinary skill in the art that (i) alternative database structures to those described may be readily employed, (ii) other memory structures besides databases may be readily employed. Any schematic illustrations and accompanying descriptions of any sample databases pre-

sented herein are exemplary arrangements for stored representations of information. Any number of other arrangements may be employed besides those suggested by the tables shown. Similarly, any illustrated entries of the databases represent exemplary information only; those skilled in the art will understand that the number and content of the entries can be different from those illustrated herein. Further, despite any depiction of the databases as tables, an object-based model could be used to store and manipulate the data types of the present invention and likewise, object methods or behaviors can be used to implement the processes of the present invention.

[0086] A “computer system” may refer to a system having one or more computers, where each computer may include a computer-readable medium embodying software to operate the computer or one or more of its components. Examples of a computer system may include: a distributed computer system for processing information via computer systems linked by a network; two or more computer systems connected together via a network for transmitting and/or receiving information between the computer systems; a computer system including two or more processors within a single computer; and one or more apparatuses and/or one or more systems that may accept data, may process data in accordance with one or more stored software programs, may generate results, and typically may include input, output, storage, arithmetic, logic, and control units.

[0087] A “network” may refer to a number of computers and associated devices that may be connected by communication facilities. A network may involve permanent connections such as cables or temporary connections such as those made through telephone or other communication links. A network may further include hard-wired connections (e.g., coaxial cable, twisted pair, optical fiber, waveguides, etc.) and/or wireless connections (e.g., radio frequency waveforms, free-space optical waveforms, acoustic waveforms, etc.). Examples of a network may include: an internet, such as the Internet; an intranet; a local area network (LAN); a wide area network (WAN); and a combination of networks, such as an internet and an intranet.

[0088] As used herein, the “client-side” application should be broadly construed to refer to an application, a page associated with that application, or some other resource or function invoked by a client-side request to the application. A “browser” as used herein is not intended to refer to any specific browser (e.g., Internet Explorer, Safari, FireFox, or the like), but should be broadly construed to refer to any client-side rendering engine that can access and display Internet-accessible resources. A “rich” client typically refers to a non-HTTP based client-side application, such as an SSH or CFIS client. Further, while typically the client-server interactions occur using HTTP, this is not a limitation either. The client server interaction may be formatted to conform to the Simple Object Access Protocol (SOAP) and travel over HTTP (over the public Internet), FTP, or any other reliable transport mechanism (such as IBM® MQSeries® technologies and CORBA, for transport over an enterprise intranet) may be used. Any application or functionality described herein may be implemented as native code, by providing hooks into another application, by facilitating use of the mechanism as a plug-in, by linking to the mechanism, and the like.

[0089] Exemplary networks may operate with any of a number of protocols, such as Internet protocol (IP), asyn-

chronous transfer mode (ATM), and/or synchronous optical network (SONET), user datagram protocol (UDP), IEEE 802.x, etc.

[0090] Embodiments of the present invention may include apparatuses for performing the operations disclosed herein. An apparatus may be specially constructed for the desired purposes, or it may comprise a general-purpose device selectively activated or reconfigured by a program stored in the device.

[0091] Embodiments of the invention may also be implemented in one or a combination of hardware, firmware, and software. They may be implemented as instructions stored on a machine-readable medium, which may be read and executed by a computing platform to perform the operations described herein.

[0092] More specifically, as will be appreciated by one skilled in the art, aspects of the present invention may be embodied as a system, method or computer program product. Accordingly, aspects of the present invention may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.” Furthermore, aspects of the present invention may take the form of a computer program product embodied in one or more computer readable medium(s) having computer readable program code embodied thereon.

[0093] In the following description and claims, the terms “computer program medium” and “computer readable medium” may be used to generally refer to media such as, but not limited to, removable storage drives, a hard disk installed in hard disk drive, and the like. These computer program products may provide software to a computer system. Embodiments of the invention may be directed to such computer program products.

[0094] An algorithm is here, and generally, considered to be a self-consistent sequence of acts or operations leading to a desired result. These include physical manipulations of physical quantities. Usually, though not necessarily, these quantities take the form of electrical or magnetic signals capable of being stored, transferred, combined, compared, and otherwise manipulated. It has proven convenient at times, principally for reasons of common usage, to refer to these signals as bits, values, elements, symbols, characters, terms, numbers or the like. It should be understood, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities.

[0095] Unless specifically stated otherwise, and as may be apparent from the following description and claims, it should be appreciated that throughout the specification descriptions utilizing terms such as “processing,” “computing,” “calculating,” “determining,” or the like, refer to the action and/or processes of a computer or computing system, or similar electronic computing device, that manipulate and/or transform data represented as physical, such as electronic, quantities within the computing system’s registers and/or memories into other data similarly represented as physical quantities within the computing system’s memories, registers or other such information storage, transmission or display devices.

[0096] Additionally, the phrase “configured to” or “operable for” can include generic structure (e.g., generic cir-

cuitry) that is manipulated by software and/or firmware (e.g., an FPGA or a general-purpose processor executing software) to operate in a manner that is capable of performing the task(s) at issue. “Configured to” may also include adapting a manufacturing process (e.g., a semiconductor fabrication facility) to fabricate devices (e.g., integrated circuits) that are adapted to implement or perform one or more tasks.

[0097] In a similar manner, the term “processor” may refer to any device or portion of a device that processes electronic data from registers and/or memory to transform that electronic data into other electronic data that may be stored in registers and/or memory. A “computing platform” may comprise one or more processors.

[0098] Embodiments within the scope of the present disclosure may also include tangible and/or non-transitory computer-readable storage media for carrying or having computer-executable instructions or data structures stored thereon. Such non-transitory computer-readable storage media can be any available media that can be accessed by a general purpose or special purpose computer, including the functional design of any special purpose processor as discussed above. By way of example, and not limitation, such non-transitory computer-readable media can include RAM, ROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code means in the form of computer-executable instructions, data structures, or processor chip design. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or combination thereof) to a computer, the computer properly views the connection as a computer-readable medium. Thus, any such connection is properly termed a computer-readable medium. Combinations of the above should also be included within the scope of the computer-readable media.

[0099] While a non-transitory computer readable medium includes, but is not limited to, a hard drive, compact disc, flash memory, volatile memory, random access memory, magnetic memory, optical memory, semiconductor based memory, phase change memory, optical memory, periodically refreshed memory, and the like; the non-transitory computer readable medium, however, does not include a pure transitory signal per se; i.e., where the medium itself is transitory.

[0100] Target multi-user computer application or target computer application may refer to any computer application which can support multiple users. Examples of such applications are Google email, Facebook and many other computer applications.

[0101] FIG. 1 is an illustration of an exemplary computer system, in accordance with an embodiment of the present invention. In the present embodiment, a user **110** typically enters data on a computing device **111**, which includes without limitation, a personal computer, laptop, Mac, cell phone, smart phone, tablet, personal digital assistant, (PDA). Computing device **111** is typically connected to a login management server **140** over a computer network **115**, which includes without limitation the Internet, a local area network (LAN), a wide area network (WAN), Bluetooth, or FireWire. The user **110** typically accesses login management server **140** to access target multi-user computer applications **130** over computer network **115**. The target multi-user

computer applications are external applications whose log-ins and passwords we are managing. The user 110 typically provides an initial login information and password, for the target multi-user computer applications 130, which may be stored in an authentication database 150, for future update and retrieval via a second network 135. It is contemplated that a password generation server 120 typically connects to target multi-user computer applications 130 to generate new passwords and update them frequently, which may occur, without limitation, hourly, daily, or some other set time interval. In the present embodiment, it is contemplated that other basis to generate new passwords include, without limitation, when the user 110 logs in or out of the application, when a user 110 is informed that his or her account has been hacked, actual request to change the password by the user 110, an important external event such as banking operations, and any other set of rules based event.

[0102] In the present embodiment, a password-generating server 120 typically interacts with the target multi-user computer applications 130 over computer network 115. A log-in instructions database 160 is typically connected to login management server 140 and the password-generating server 120 over second network 135. Log-in instructions database 160 typically stores instructions for logging in, logging out, or changing passwords for the target multi-user computer applications 130. Password-generating server 120 typically connects to log-in instructions database 160 to obtain instructions for logging in, logging out, and updating passwords on target multi-user computer applications server 130. Password-generating server 120 typically retrieves the current password, for the target multi-user computer applications 130, from authentication database 150 and updates it, once a new password has been generated and the target multi-user computer applications have been updated to this new password, by the password generating server 120, following the login and password update instructions for the target applications 130, stored in Login Instructions Database 160. In an alternative embodiment, password-generating server 120 may not exist and a software implementation on login management server 140 that performs all the functions of password-generating server 120 is possible.

[0103] In an alternative embodiment, log-in instructions database 160 may be incorporated in the same authentication database 150.

[0104] FIG. 2 is a flowchart of an exemplary method to frequently change a user's password for target multi-user computer applications, in accordance with an embodiment of the present invention. The password generation server 120, runs in a continuous loop in the background, checking whether the current time is such that it has exceeded or equals the time specified by the frequency at which the given user's password should be changed, for the target multi-user computer applications 130, at step 215. It is contemplated that when continuous password changes are desired, the time interval will typically be zero. If the time limit has not been exceeded, the password generation server 120 will wait at step 220. If the time limit has been exceeded, then the password generation server 120, may access login instructions database 160, to change user's 110 password for the given target computer application at step 225. Password generation server 120 also accesses the login instructions database 160 to enable password generation server 120 to login and logout from the target multi-user computer applications server 130 as the given user at step 230. Password

generation server 120 then generates a new random password at step 235. The new password is typically stored on authentication database 150 at step 240. Password generation server 120 executes instructions to login to the target computer application server 130 with the identity of user 110 at step 245. Password generation server 120 applies the new password generated from step 235 to change user's 110 password at step 250. Password generation server 120 ends the session by logging out of target multi-user computer applications 130 at step 255. It is contemplated the process completes at step 260.

[0105] In the present embodiment, an exemplary application of the login aspect may include, without limitation, computer code in the Java language, which utilizes the in-built HTTP Client class in Java by executing the HTTP GET and POST requests, to login, update passwords, and logout. The implementable instructions for these steps would be coded using the XML language, which can be read by Java. Similarly, other languages such as JavaScript, or any other computer language may be used to develop the computer code necessary to execute these steps.

[0106] FIG. 3 is a flowchart of an exemplary method for changing a user's password for all users and all registered target multi-user computer applications, in accordance with an embodiment of the present invention. In the present embodiment, password generation server 120 accesses authentication database 150 at step 305. It is contemplated that password generation server 120 checks all users available at step 310. Password generation server 120 may continuously cycle through a loop wherein it reads the list of all registered users and typically begins the loop for each user at step 315. Password generation server 120 typically reads the set of target multi-user computer applications 130 for which a particular user has a login ID and password at step 320. Password generation server 120, typically checks each application in this list at step 325. Password generation server 120 changes the user's 110 password for all of user's 110 registered target multi-user computer applications 130, at typically regular frequent or even continuous intervals, for all registered users at step 330. It is contemplated that password generation server 120 may check whether all applications have been processed at step 335. If all applications have not been processed, password generation server 120 returns to step 325. If all applications have been processed, password generation server 120 typically checks whether all users have been processed at step 340. If all users have not been processed, password generation server 120 returns to step 315. If all users have been processed, password generation server 120 ends the loop at step 345.

[0107] FIG. 4 is a flowchart of an exemplary method to allow a user to still login to their target computer application, in accordance with an embodiment of the present invention, without having knowledge of their frequently changing passwords. In the present embodiment, the user 110 starts at login management server 140 at step 410. It is contemplated login management server 140 may perform a login on behalf of user 110 to the target application on target applications server 130 at step 420. It is contemplated that password-generating server 120 continually or frequently changes the user's passwords, at predefined intervals, and thus, user 110 is no longer in control or even necessarily aware of his or her password for the target application. The new password is stored in authentication database 150. Login management server 140 reads the user's current password from authen-

tication database **150** for target multi-user computer applications server **130** at step **430**. Login management server **140** uses the new password to login to the target computer application on the user's **110** behalf at step **440**. Once the login is complete, login management server **140** transfers the login session to the user's **110** web browser or other computer application that user **110** is currently using to access the target multi-user computer application at step **450**. One means to transfer this session, without limitation, is to transfer all the session cookies to the user's web browser, from the Login management server **140**, thus passing the control of the logged in session to the user. After login management server **140** transfers control to user's **110** browser, it is contemplated the process will end at step **460**.

[0108] FIG. **5** is a flowchart of an exemplary method to allow a user to identify a target computer application, optionally specify the login, logout, and password change instructions, as well as the initial login password, in accordance with an embodiment of the present invention. In the present embodiment, a user **110** may access login management server **140** to enter the user's **110** target multi-user computer applications, and the related login ID, passwords, and login instructions on log-in instructions database **160** at step **510**. User typically first logs into Login management server **140** at step **520**. It is contemplated that user **110** may proceed in one of two ways. In a first aspect, a user **110** selects from a list of known target systems typically located on target multi-user computer applications server **130** at step **530**. In the second aspect, the target computer application is not known, and therefore, the user **110** uploads to the login management server with instructions to login, logout and change password for the target computer application at step **540**. After this step **540**, the target applications are now known to the Login management server **140**, as well as the password generation server, **120**. It is contemplated that a target multi-user computer application may be selected after the previous step is complete. Once the target application has been selected, user **110** provides the login ID and password for login management server **140** at step **550**.

[0109] In the present embodiment, user **110** selects or provides the frequency at which to auto-change passwords for a particular target computer application. It is contemplated the frequency may be continuous, including, without limitation, from seconds, days, months, years, or any discrete time interval at step **560**. Further, without limitation, the user may select other conditions for changing the password, for example, always change password for the user **110** on the target applications **130**, every time the user logs in and out of target systems, etc. Login management server **140** stores all the login, logout, password change instructions, target system name, identifier, for the target multi-user computer applications **130**, in the login instructions database **160**, and user ID, password change frequency information and password for the target multi-user computer applications **130**, in the authentication database **150** at step **570**. Login management server **140** typically completes operations at step **580**.

[0110] FIG. **6** is an illustration of an exemplary user interface of an Internet website allowing users to login to a computer system, in accordance with an embodiment of the present invention.

[0111] FIG. **7** is an illustration of an exemplary user interface of an Internet website allowing users to access their

target multi-user computer applications with changed passwords, in accordance with an embodiment of the present invention.

[0112] FIG. **8** is an illustration of an exemplary user interface of an Internet website allowing users to register new target multi-user computer applications, in accordance with an embodiment of the present invention.

[0113] It is to be understood that any exact measurements/dimensions or particular construction materials indicated herein are solely provided as examples of suitable configurations and are not intended to be limiting in any way. Depending on the needs of the particular application, those skilled in the art will readily recognize, in light of the following teachings, a multiplicity of suitable alternative implementation details.

[0114] Those skilled in the art will readily recognize, in light of and in accordance with the teachings of the present invention, that any of the foregoing steps and/or system modules may be suitably replaced, reordered, removed and additional steps and/or system modules may be inserted depending upon the needs of the particular application, and that the systems of the foregoing embodiments may be implemented using any of a wide variety of suitable processes and system modules, and is not limited to any particular computer hardware, software, middleware, firmware, microcode and the like. For any method steps described in the present application that can be carried out on a computing machine, a typical computer system can, when appropriately configured or designed, serve as a computer system in which those aspects of the invention may be embodied.

[0115] FIG. **9** is a block diagram depicting an exemplary client/server system, which may be used by an exemplary web-enabled/networked embodiment of the present invention.

[0116] A communication system **900** includes a multiplicity of clients with a sampling of clients denoted as a client **902** and a client **904**, a multiplicity of local networks with a sampling of networks denoted as a local network **909** and a local network **908**, a global network **910** and a multiplicity of servers with a sampling of servers denoted as a server **912** and a server **914**.

[0117] Client **902** may communicate bi-directionally with local network **909** via a communication channel **919**. Client **904** may communicate bi-directionally with local network **908** via a communication channel **918**. Local network **909** may communicate bi-directionally with global network **910** via a communication channel **920**. Local network **908** may communicate bi-directionally with global network **910** via a communication channel **922**. Global network **910** may communicate bi-directionally with server **912** and server **914** via a communication channel **924**. Server **912** and server **914** may communicate bi-directionally with each other via communication channel **924**. Furthermore, clients **902**, **904**, local networks **909**, **908**, global network **910** and servers **912**, **914** may each communicate bi-directionally with each other.

[0118] In one embodiment, global network **910** may operate as the Internet. It will be understood by those skilled in the art that communication system **900** may take many different forms. Non-limiting examples of forms for communication system **900** include local area networks (LANs), wide area networks (WANs), wired telephone networks,

wireless networks, or any other network supporting data communication between respective entities.

[0119] Clients **902** and **904** may take many different forms. Non-limiting examples of clients **902** and **904** include personal computers, personal digital assistants (PDAs), cellular phones and smartphones.

[0120] Client **902** includes a CPU **929**, a pointing device **928**, a keyboard **930**, a microphone **932**, a printer **934**, a memory **939**, a mass memory storage **938**, a GUI **940**, a video camera **942**, an input/output interface **944** and a network interface **949**.

[0121] CPU **929**, pointing device **928**, keyboard **930**, microphone **932**, printer **934**, memory **939**, mass memory storage **938**, GUI **940**, video camera **942**, input/output interface **944** and network interface **949** may communicate in a unidirectional manner or a bi-directional manner with each other via a communication channel **948**. Communication channel **948** may be configured as a single communication channel or a multiplicity of communication channels.

[0122] CPU **929** may be comprised of a single processor or multiple processors. CPU **929** may be of various types including micro-controllers (e.g., with embedded RAM/ROM) and microprocessors such as programmable devices (e.g., RISC or SISC based, or CPLDs and FPGAs) and devices not capable of being programmed such as gate array ASICs (Application Specific Integrated Circuits) or general purpose microprocessors.

[0123] As is well known in the art, memory **939** is used typically to transfer data and instructions to CPU **929** in a bi-directional manner. Memory **939**, as discussed previously, may include any suitable computer-readable media, intended for data storage, such as those described above excluding any wired or wireless transmissions unless specifically noted. Mass memory storage **938** may also be coupled bi-directionally to CPU **929** and provides additional data storage capacity and may include any of the computer-readable media described above. Mass memory storage **938** may be used to store programs, data and the like and is typically a secondary storage medium such as a hard disk. It will be appreciated that the information retained within mass memory storage **938**, may, in appropriate cases, be incorporated in standard fashion as part of memory **939** as virtual memory.

[0124] CPU **929** may be coupled to GUI **940**. GUI **940** enables a user to view the operation of computer operating system and software. CPU **929** may be coupled to pointing device **928**. Non-limiting examples of pointing device **928** include computer mouse, trackball and touchpad. Pointing device **928** enables a user with the capability to maneuver a computer cursor about the viewing area of GUI **940** and select areas or features in the viewing area of GUI **940**. CPU **929** may be coupled to keyboard **930**. Keyboard **930** enables a user with the capability to input alphanumeric textual information to CPU **929**. CPU **929** may be coupled to microphone **932**. Microphone **932** enables audio produced by a user to be recorded, processed and communicated by CPU **929**. CPU **929** may be connected to printer **934**. Printer **934** enables a user with the capability to print information to a sheet of paper. CPU **929** may be connected to video camera **942**. Video camera **942** enables video produced or captured by user to be recorded, processed and communicated by CPU **929**.

[0125] CPU **929** may also be coupled to input/output interface **944** that connects to one or more input/output

devices such as such as CD-ROM, video monitors, track balls, mice, keyboards, microphones, touch-sensitive displays, transducer card readers, magnetic or paper tape readers, tablets, styluses, voice or handwriting recognizers, or other well-known input devices such as, of course, other computers.

[0126] Finally, CPU **929** optionally may be coupled to network interface **949** which enables communication with an external device such as a database or a computer or telecommunications or internet network using an external connection shown generally as communication channel **919**, which may be implemented as a hardwired or wireless communications link using suitable conventional technologies. With such a connection, CPU **929** might receive information from the network, or might output information to a network in the course of performing the method steps described in the teachings of the present invention. FIG. **10** illustrates a block diagram depicting an exemplary regionalized client/server communication system supporting frequent password generation for target multi-user computer applications, in accordance with an embodiment of the present invention.

[0127] A computer system **1000** includes users **1002** accessing a login management server **1008**, a target multi-user computer applications server **1004**, password generating server **1010**, authentication database **1012**, and log-in instructions database **1014**. Users **1002** provide login and logout information **1006**.

[0128] Users **1002** access login management server **1008** to input login and logout information **1006** to access target multi-user computer applications server **1004**. Password-generating server **1010**, thereafter, generates a new random password for target multi-user computer applications, at desired time intervals or events. The new passwords are stored in authentication database **1012**, while log-in instructions for each target computer application are stored in log-in instructions database **1014**.

[0129] In some implementations, system **1000** may operate as the Internet. It will be understood by those skilled in the art that communication system **1000** may take many different forms. Non-limiting examples of forms for computer system **1000** include local area networks (LANs), wide area networks (WANs), wired telephone networks, cellular telephone networks or any other network supporting data communication between respective entities via hardwired or wireless communication networks. Authenticating server **710** may operate to transfer information between the various networked elements.

[0130] Login management server **1008** may operate to execute software instructions, store information and communicate with other networked elements. Non-limiting examples of software and scripting languages which may be executed on login management server **1008** include C, C++, C# and Java.

[0131] Users **1002** may operate to communicate bi-directionally with login management server **1008**. Login management server **1008** may operate to communicate bi-directionally with password-generating server **1010**, authentication database **1012**, or log-in instructions database **1014**.

[0132] FIG. **11** illustrates a block diagram depicting a conventional client/server communication system.

[0133] A communication system **1100** includes a multiplicity of networked regions with a sampling of regions

denoted as a network region **1102** and a network region **1104**, a global network **1106** and a multiplicity of servers with a sampling of servers denoted as a server device **1108** and a server device **1110**.

[0134] Network region **1102** and network region **1104** may operate to represent a network contained within a geographical area or region. Non-limiting examples of representations for the geographical areas for the networked regions may include postal zip codes, telephone area codes, states, counties, cities and countries. Elements within network region **1102** and **1104** may operate to communicate with external elements within other networked regions or within elements contained within the same network region.

[0135] In some implementations, global network **1106** may operate as the Internet. It will be understood by those skilled in the art that communication system **1100** may take many different forms. Non-limiting examples of forms for communication system **1100** include local area networks (LANs), wide area networks (WANs), wired telephone networks, cellular telephone networks or any other network supporting data communication between respective entities via hardwired or wireless communication networks. Global network **1106** may operate to transfer information between the various networked elements.

[0136] Server device **1108** and server device **1110** may operate to execute software instructions, store information, support database operations and communicate with other networked elements. Non-limiting examples of software and scripting languages which may be executed on server device **1108** and server device **1110** include C, C++, C# and Java.

[0137] Network region **1102** may operate to communicate bi-directionally with global network **1106** via a communication channel **1112**. Network region **1104** may operate to communicate bi-directionally with global network **1106** via a communication channel **1114**. Server device **1108** may operate to communicate bi-directionally with global network **1106** via a communication channel **1116**. Server device **1110** may operate to communicate bi-directionally with global network **1106** via a communication channel **1118**. Network region **1102** and **1104**, global network **1106** and server devices **1108** and **1110** may operate to communicate with each other and with every other networked device located within communication system **1100**.

[0138] Server device **1108** includes a networking device **1120** and a server **1122**. Networking device **1120** may operate to communicate bi-directionally with global network **1106** via communication channel **1116** and with server **1122** via a communication channel **1124**. Server **1122** may operate to execute software instructions and store information.

[0139] Network region **1102** includes a multiplicity of clients with a sampling denoted as a client **1126** and a client **1128**. Client **1126** includes a networking device **1134**, a processor **1136**, a GUI **1138** and an interface device **1140**. Non-limiting examples of devices for GUI **1138** include monitors, televisions, cellular telephones, smartphones and PDAs (Personal Digital Assistants). Non-limiting examples of interface device **1140** include pointing device, mouse, trackball, scanner and printer. Networking device **1134** may communicate bi-directionally with global network **1106** via communication channel **1112** and with processor **1136** via a communication channel **1142**. GUI **1138** may receive information from processor **1136** via a communication channel **1144** for presentation to a user for viewing. Interface device

1140 may operate to send control information to processor **1136** and to receive information from processor **1136** via a communication channel **1146**. Network region **1104** includes a multiplicity of clients with a sampling denoted as a client **1130** and a client **1132**. Client **1130** includes a networking device **1148**, a processor **1150**, a GUI **1152** and an interface device **1154**. Non-limiting examples of devices for GUI **1138** include monitors, televisions, cellular telephones, smartphones and PDAs (Personal Digital Assistants). Non-limiting examples of interface device **1140** include pointing devices, mouse, trackballs, scanners and printers. Networking device **1148** may communicate bi-directionally with global network **1106** via communication channel **1114** and with processor **1150** via a communication channel **1156**. GUI **1152** may receive information from processor **1150** via a communication channel **1158** for presentation to a user for viewing. Interface device **1154** may operate to send control information to processor **1150** and to receive information from processor **1150** via a communication channel **1160**.

[0140] For example, consider the case where a user interfacing with client **1126** may want to execute a networked application. A user may enter the IP (Internet Protocol) address for the networked application using interface device **1140**. The IP address information may be communicated to processor **1136** via communication channel **1146**. Processor **1136** may then communicate the IP address information to networking device **1134** via communication channel **1142**. Networking device **1134** may then communicate the IP address information to global network **1106** via communication channel **1112**. Global network **1106** may then communicate the IP address information to networking device **1120** of server device **1108** via communication channel **1116**. Networking device **1120** may then communicate the IP address information to server **1122** via communication channel **1124**. Server **1122** may receive the IP address information and after processing the IP address information may communicate return information to networking device **1120** via communication channel **1124**. Networking device **1120** may communicate the return information to global network **1106** via communication channel **1116**. Global network **1106** may communicate the return information to networking device **1134** via communication channel **1112**. Networking device **1134** may communicate the return information to processor **1136** via communication channel **1142**. Processor **1136** may communicate the return information to GUI **1138** via communication channel **1144**. User may then view the return information on GUI **1138**.

[0141] It will be further apparent to those skilled in the art that at least a portion of the novel method steps and/or system components of the present invention may be practiced and/or located in location(s) possibly outside the jurisdiction of the United States of America (USA), whereby it will be accordingly readily recognized that at least a subset of the novel method steps and/or system components in the foregoing embodiments must be practiced within the jurisdiction of the USA for the benefit of an entity therein or to achieve an object of the present invention. Thus, some alternate embodiments of the present invention may be configured to comprise a smaller subset of the foregoing means for and/or steps described that the applications designer will selectively decide, depending upon the practical considerations of the particular implementation, to carry out and/or locate within the jurisdiction of the USA.

For example, any of the foregoing described method steps and/or system components which may be performed remotely over a network (e.g., without limitation, a remotely located server) may be performed and/or located outside of the jurisdiction of the USA while the remaining method steps and/or system components (e.g., without limitation, a locally located client) of the foregoing embodiments are typically required to be located/performed in the USA for practical considerations. In client-server architectures, a remotely located server typically generates and transmits required information to a US based client, for use according to the teachings of the present invention. Depending upon the needs of the particular application, it will be readily apparent to those skilled in the art, in light of the teachings of the present invention, which aspects of the present invention can or should be located locally and which can or should be located remotely. Thus, for any claims construction of the following claim limitations that are construed under 35 USC §112 (6) it is intended that the corresponding means for and/or steps for carrying out the claimed function are the ones that are locally implemented within the jurisdiction of the USA, while the remaining aspect(s) performed or located remotely outside the USA are not intended to be construed under 35 USC §112 (6). In some embodiments, the methods and/or system components which may be located and/or performed remotely include, without limitation: The entire solution may be hosted outside the US for non-US clients and users **110**.

[0142] It is noted that according to USA law, all claims must be set forth as a coherent, cooperating set of limitations that work in functional combination to achieve a useful result as a whole. Accordingly, for any claim having functional limitations interpreted under 35 USC §112 (6) where the embodiment in question is implemented as a client-server system with a remote server located outside of the USA, each such recited function is intended to mean the function of combining, in a logical manner, the information of that claim limitation with at least one other limitation of the claim. For example, in client-server systems where certain information claimed under 35 USC §112 (6) is/(are) dependent on one or more remote servers located outside the USA, it is intended that each such recited function under 35 USC §112 (6) is to be interpreted as the function of the local system receiving the remotely generated information required by a locally implemented claim limitation, wherein the structures and or steps which enable, and breath life into the expression of such functions claimed under 35 USC §112 (6) are the corresponding steps and/or means located within the jurisdiction of the USA that receive and deliver that information to the client (e.g., without limitation, client-side processing and transmission networks in the USA). When this application is prosecuted or patented under a jurisdiction other than the USA, then "USA" in the foregoing should be replaced with the pertinent country or countries or legal organization(s) having enforceable patent infringement jurisdiction over the present application, and "35 USC §112 (6)" should be replaced with the closest corresponding statute in the patent laws of such pertinent country or countries or legal organization(s).

[0143] Alternative implementations which bifurcate or combine the various components are still the same system. For example, the login management server **140**, the password generation server **120**, the authentication database **150**, the login instructions database **160** could all be combined

into a single physical server, but would logically be still described by the system listed in this invention. On the other extreme, the various components could be hosted or owned by different corporations, for example the login management server **140** could be owned by a different company or could be subdivided further, same for password generation server **120** and the two databases **150** and **160** but logically they would be implementing the same system as described in this patent application.

[0144] All such alternatives should be considered within the purview of this application.

[0145] Indeed the authors of this invention have created a security management appliance which combines the login management server **140**, the password generation server **120**, the authentication database **150**, and the login instructions database **160** into a single physical computer appliance.

[0146] All the features disclosed in this specification, including any accompanying abstract and drawings, may be replaced by alternative features serving the same, equivalent or similar purpose, unless expressly stated otherwise. Thus, unless expressly stated otherwise, each feature disclosed is one example only of a generic series of equivalent or similar features.

[0147] It is noted that according to USA law 35 USC §112 (1), all claims must be supported by sufficient disclosure in the present patent specification, and any material known to those skilled in the art need not be explicitly disclosed. However, 35 USC §112 (6) requires that structures corresponding to functional limitations interpreted under 35 USC §112 (6) must be explicitly disclosed in the patent specification. Moreover, the USPTO's Examination policy of initially treating and searching prior art under the broadest interpretation of a "mean for" claim limitation implies that the broadest initial search on 112(6) functional limitation would have to be conducted to support a legally valid Examination on that USPTO policy for broadest interpretation of "mean for" claims. Accordingly, the USPTO will have discovered a multiplicity of prior art documents including disclosure of specific structures and elements which are suitable to act as corresponding structures to satisfy all functional limitations in the below claims that are interpreted under 35 USC §112 (6) when such corresponding structures are not explicitly disclosed in the foregoing patent specification. Therefore, for any invention element(s)/structure(s) corresponding to functional claim limitation(s), in the below claims interpreted under 35 USC §112 (6), which is/are not explicitly disclosed in the foregoing patent specification, yet do exist in the patent and/or non-patent documents found during the course of USPTO searching, Applicant(s) incorporate all such functionally corresponding structures and related enabling material herein by reference for the purpose of providing explicit structures that implement the functional means claimed. Applicant(s) request(s) that fact finders during any claims construction proceedings and/or examination of patent allowability properly identify and incorporate only the portions of each of these documents discovered during the broadest interpretation search of 35 USC §112 (6) limitation, which exist in at least one of the patent and/or non-patent documents found during the course of normal USPTO searching and or supplied to the USPTO during prosecution. Applicant(s) also incorporate by reference the bibliographic citation information to identify all such documents comprising functionally corresponding

structures and related enabling material as listed in any PTO Form-892 or likewise any information disclosure statements (IDS) entered into the present patent application by the USPTO or Applicant(s) or any 3rd parties. Applicant(s) also reserve its right to later amend the present application to explicitly include citations to such documents and/or explicitly include the functionally corresponding structures which were incorporate by reference above.

[0148] Thus, for any invention element(s)/structure(s) corresponding to functional claim limitation(s), in the below claims, that are interpreted under 35 USC §112 (6), which is/are not explicitly disclosed in the foregoing patent specification, Applicant(s) have explicitly prescribed which documents and material to include the otherwise missing disclosure, and have prescribed exactly which portions of such patent and/or non-patent documents should be incorporated by such reference for the purpose of satisfying the disclosure requirements of 35 USC §112 (6). Applicant(s) note that all the identified documents above which are incorporated by reference to satisfy 35 USC §112 (6) necessarily have a filing and/or publication date prior to that of the instant application, and thus are valid prior documents to incorporated by reference in the instant application.

[0149] Having fully described at least one embodiment of the present invention, other equivalent or alternative methods of implementing continuously or frequently changing users' passwords for external multi-user computer applications, while still allowing the users to transparently access these external multi-user applications according to the present invention will be apparent to those skilled in the art. Various aspects of the invention have been described above by way of illustration, and the specific embodiments disclosed are not intended to limit the invention to the particular forms disclosed. The particular implementation of the continuously or frequently changing users' passwords for external multi-user computer applications, while still allowing the users to transparently access these external multi-user applications may vary depending upon the particular context or application. By way of example, and not limitation, the continuously or frequently changing users' passwords for external multi-user computer applications, while still allowing the users to transparently access these external multi-user applications described in the foregoing were principally directed to frequent random password generation implementations; however, similar techniques may instead be applied to random biometric inputs based on repeated scans and adding variations that are known only to the computer system, which implementations of the present invention are contemplated as within the scope of the present invention. The invention is thus to cover all modifications, equivalents, and alternatives falling within the spirit and scope of the following claims. It is to be further understood that not all of the disclosed embodiments in the foregoing specification will necessarily satisfy or achieve each of the objects, advantages, or improvements described in the foregoing specification.

[0150] Claim elements and steps herein may have been numbered and/or lettered solely as an aid in readability and understanding. Any such numbering and lettering in itself is not intended to and should not be taken to indicate the ordering of elements and/or steps in the claims.

[0151] The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material,

or act for performing the function in combination with other claimed elements as specifically claimed.

[0152] The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present invention has been presented for purposes of illustration and description, but is not intended to be exhaustive or limited to the invention in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the invention. The embodiment was chosen and described in order to best explain the principles of the invention and the practical application, and to enable others of ordinary skill in the art to understand the invention for various embodiments with various modifications as are suited to the particular use contemplated.

[0153] The Abstract is provided to comply with 37 C.F.R. Section 1.72(b) requiring an abstract that will allow the reader to ascertain the nature and gist of the technical disclosure. That is, the Abstract is provided merely to introduce certain concepts and not to identify any key or essential features of the claimed subject matter. It is submitted with the understanding that it will not be used to limit or interpret the scope or meaning of the claims.

[0154] The following claims are hereby incorporated into the detailed description, with each claim standing on its own as a separate embodiment.

What is claimed is:

1. A computerized method comprising the steps of:
 - connecting a computing device to a login management server over a computer network;
 - entering at least a user login information and a first password on said computing device;
 - storing said user login information and said first password in an authentication database for at least updating and retrieval of said first password via said computer network;
 - accessing at least one target multi-user computer application stored on a target multi-user computer applications server through said login management server over said computer network;
 - connecting a login instructions database to said login management server;
 - connecting a password generation server to said login instructions database; and
 - obtaining instructions, by said password generation server from said login instructions database, wherein said instructions comprising at least a logging in instruction, a logging out instruction, and an updating instruction of at least one utilized password on said target multi-user computer applications server.
2. The method of claim 1, further comprising the step of connecting said password generation server to said target multi-user computer applications server configured to perform a password update.
3. The method of claim 2, further comprising the step of generating, by said password generation server, a second password.
4. The method of claim 3, further comprising the step of updating said first password with said generated second password.

5. The method of claim 4, wherein said password update is an automatic password update based on a preset time interval.

6. The method of claim 5, wherein the user is unaware of said automatic password update and without disrupting the user's access to said target multi-user computer applications.

7. The method of claim 4, in which said password update comprises said password generation server executing instructions to login to said target computer application server with the identity of the user.

8. The method of claim 7, wherein said password update further comprises said password generation server applying said generated second password without the knowledge of the user.

9. The method of claim 2, further comprising the step of interacting with said target multi-user computer applications server over said computer network.

10. The method of claim 9, further comprising the step of retrieving said first password, for the target multi-user computer applications, from said authentication database.

11. The method of claim 10, further comprising the step of applying said password update, once a second password has been generated and the target multi-user computer applications have been updated to said second password, by said password generating server, following said login and password update instructions for said target applications, stored in said login instructions database.

12. A system comprising:

means for entering at least a user login information and a first password;

means for storing said user login information and said first password;

means for storing at least one target multi-user computer application;

means for providing logging in, logging out, and updating instructions;

means for generating random passwords; and

means for updating said first password with a generated second random password.

13. The method of claim 12, wherein said password update is an automatic password update based on a preset time interval.

14. The method of claim 13, wherein the user is unaware of said automatic password update and without disrupting the user's access to said target multi-user computer application.

15. The method of claim 12, in which said password update comprises said password generating means executing said instructions to login to said target multi-user computer application storing means with an identity of the user.

16. The method of claim 15, wherein said password generating means applying said generated random password without the knowledge of the user.

17. A computerized method comprising the steps of:

connecting a computing device to a login management server over a first computer network;

entering at least a user login information and a first password on said computing device;

storing said user login information and said first password in an authentication database for a predetermined retrieval and updating of said first password;

accessing at least one target multi-user computer application stored on a target multi-user computer applications server through said login management server over said first computer network;

connecting a login instructions database to said login management server;

connecting a password generation server to said login instructions database; and

obtaining instructions, by said password generation server from said login instructions database, wherein said instructions comprising at least one of a logging in instruction, a logging out instruction, and an updating instruction of at least one password on said target multi-user computer applications server.

18. The method of claim 17, further comprising the step of retrieving said first password, for the target multi-user computer applications, from said authentication database.

19. The method of claim 18, in which said password update comprises said password generation server executing instructions to login to said target computer application server with the identity of the user.

20. The method of claim 19, wherein said password update further comprises said password generation server applying said generated second password automatically without the knowledge of the user.

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